Technical Writing for Engineers & Scientists

Fourth Edition

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Mc Graw Hill

TECHNICAL WRITING FOR ENGINEERS & SCIENTISTS, FOURTH EDITION

This book is printed on acid-free paper.

1 2 3 4 5 6 7 8 9 LWI 27 26 25 24 23 22

ISBN 978-0-073-53493-0 (bound edition)
MHID 0-073-53493-5 (bound edition)
ISBN 978-1-264-16315-1 (loose-leaf edition)
MHID 1-264-16315-0 (loose-leaf edition)

Portfolio Manager: *Beth Bettcher*
Product Developer: *Erin Kamm*
Marketing Manager: *Shannon O'Donnell*
Content Project Managers: *Maria McGreal, Tammy Juran*
Buyer: *Rachel Hirschfield*
Content Licensing Specialist: *Beth Cray*
Cover Image: *Leslie A. Potter*
Compositor: *Aptara®, Inc.*

Library of Congress Cataloging-in-Publication Data

Names: Finkelstein, Leo, 1946- author. | Aune, Jeanine Elise, author. | Potter, Leslie A., author.
Title: Technical writing for engineers & scientists / Leo Finkelstein, Jr., Wright State University, Jeanine Elise Aune, Iowa State University, Leslie A. Potter, Iowa State University.
Other titles: Pocket book of technical writing for engineers and scientists | Technical writing for engineers and scientists
Description: Fourth edition. | Dubuque : McGraw Hill Education, 2022. | Includes index.
Identifiers: LCCN 2021031273 (print) | LCCN 2021031274 (ebook) | ISBN 9780073534930 (acid-free paper) | ISBN 9781264163151 (spiral bound) | ISBN 9781264163106 (ebook) | ISBN 9780077591724 (ebook other)
Subjects: LCSH: Technical writing.
Classification: LCC T11 .F53 2022 (print) | LCC T11 (ebook) | DDC 808.06/66–dc23
LC record available at https://lccn.loc.gov/2021031273
LC ebook record available at https://lccn.loc.gov/2021031274

mheducation.com/highered

Dedication

In memory of Dr. Leo Finkelstein, Jr. We are honored to continue his legacy.

This has been one of my favorite projects of all time due, in no small part, to the support of and collaboration with some of my favorite people. It is dedicated to Siggi for unwavering support, to my children for just being, to my pets and pony for the cuddles and chances to recharge my soul, to my mom for being my cheerleader, to my siblings for their wit and humor, and to the legendary Grandma Glenna for showing me the way to teaching and about not letting one's standards slip. And last but not least, to Leslie, for her creativity, enthusiasm, and for keeping my nose to the grindstone.

–Jenny

I have always loved language: it was difficult for me to choose between a career in engineering and a career in English. I am grateful for the opportunity to combine both interests in this book and couldn't have asked for a better writing partner–thank you, Jenny! I dedicate it to John, for all that he has done these past 30 years to make it possible for me to tackle this project; as well as to our three boys for their never-ending good humor, generosity of spirit, and willing assistance; to my parents for giving me the best of each of themselves, including my mom's amazing editing skills and my dad's spot-on advice; to my brother who supports me always; and to my in-laws whom I love dearly. Shout-out to my dogs, too, who keep me grounded.

–Leslie

Preface

Purpose

The purpose of this book is to succinctly explain the content and structure of concepts and genres common to communication in engineering and science disciplines. Much like Dr. Finkelstein did in the first three editions, we aim to avoid "the sterile, encyclopedic treatment of writing concepts" that exists in many textbooks about writing. While such textbooks can be helpful for writing instructors who want to cover all of the ins and outs of technical writing theory, concepts, strategies, and genres in writing classes, such comprehensive textbooks might not be the most useful for instructors looking to incorporate writing assignments into their already-packed classes, or for students looking for the nitty-gritty details about what they need to do to get the writing project done in their engineering and science classes.

Approach

Our approach to revising this textbook was based on our combined 50+ years of teaching experience. We have endeavored to bring our approach to teaching to *Technical Writing for Engineers and Scientists*, 4th edition.

Theoretical Foundation

Technical communication is most effective when it considers audience, purpose, and context. Audiences can be categorized in many ways, but one of the most utilitarian methods is to think of them as decision-makers, advisors, and implementors. For example, when you are writing an abstract or summary, you are typically writing for decision-makers and/or advisors. When you are researching and writing a feasibility or recommendation report, you are writing for decision-makers and advisors. When you are writing descriptions or instructions, you are most likely writing for implementors. We have considered these three categories of audience as we have revised the content in this book.

In addition to audience, technical communication is most effective when it considers purpose and context. If we have been hired by the CEO and founder of a major pasta producer to write a report on the feasibility of moving a pasta factory to the upper Midwest, and we write a description of a pasta factory and its components, we will have singularly failed in understanding our purpose. The CEO is already familiar with a pasta factory; they need an evaluation of a solution based on a set of criteria, like proximity to rail lines for shipping raw ingredients. As technical writers, we must anticipate how our communication will be used and in what context. For example, electronic instructions for an executive in their corner office have an entirely different context than those required by a worker standing underneath a molten iron transfer line in a foundry.

Restructured for Easier Understanding

Over the years, we have learned that our students do best when they can see a finished example before we get into the details—much like assembly instructions and recipes, it helps the implementor to look at a picture of the final product before they begin crafting it. Therefore, we restructured the textbook's genre chapter content to provide a definition first, followed by an overview, a basic general outline, a complete example, that example broken down into the logical moves that the writer needs to make, and if relevant, additional examples illustrating the range of that technical writing genre. The intent is that students see the overall document to "get a feel" for it before they examine the breakdown of the components in that example document.

We have also incorporated references to other chapters along with small excerpts of copied-and-pasted text throughout most of the book. We did this to facilitate the use of individual chapters rather than expecting a student to read and remember the entire book. Bonus: repetition is how we move knowledge into our long-term memory.

Analogies

Over our years of teaching, we have developed the tendency to explain new concepts and ideas to students using analogies (lightbulb moments!). In this revision, we also tried to connect technical writing concepts and genres to a framework with which (we hope!) students are familiar. While not everyone loves dogs, and some may be allergic or even avoid canines for religious reasons, we anticipate that most everyone will at least be familiar with the concept of the dog, its multiple variations and roles in society, and purposes behind those variations. We hope that our readers find the analogies helpful, if somewhat wacky, in learning about technical writing concepts and genres.

Although we do assume that most everyone is familiar with dogs in general, we do not assume that everyone knows all of the various dog species by name, and here we took our own advice from Chapter 4 on Technical Definitions:

> **In some situations, you might need to sacrifice desired precision in your definition to achieve the required level of communication.**

To ensure that our readers clearly recognize our references to specific dog breeds and their connection to genres, we have treated dog breed names as proper nouns despite generally accepted capitalization guidelines for dog breeds. For example, in most writing situations, "border collie" would not be capitalized, nor would the "pinscher" in "Doberman pinscher." However, we have capitalized all words in each dog breed name so that readers will know we are talking about Border Collies and Doberman Pinschers as dog genres, analogous to writing genres.

Embraced Our Inner Goofy

Finally, we have tried to incorporate the same Goofy (dog pun intended) sense of humor that we try to share in our courses to make writing as fun and interesting for our students as we can. It was Finkelstein's light-heartedness and willingness to poke fun at himself that initially attracted us to his textbook, and we are more than happy to continue the tradition.

Disclaimers

As part of our attempts to be light-hearted, we have used numerous fictitious names in examples throughout the book. Any similarity to actual humans, towns, or organizations is completely coincidental.

Also, just as many sources for students writing technical reports have moved online in the past decade, so have they migrated online for authors writing textbooks. We have cited numerous websites throughout the text, including access dates, but understand that these addresses might change over time. Our intent was to provide enough information for readers to be able to search the topics successfully even if the websites change.

Organization

We organized the fourth edition around three major sections: the first one discusses fundamental material, the second describes how to write the most common technical documents, and the third provides useful information that, frankly, does not fit neatly in the first two sections.

Section I: Fundamentals

Chapters 1 through 6 deal with basic considerations, including the component skills you will need to produce effective technical writing. Expanding on the successful approach used in the first three editions, this section includes

- **Chapter 1: Introduction** explains what technical writing is and the basic concepts needed in technical writing.
- **Chapter 2: Ethical Considerations** focuses on ethics in technical writing and includes a general discussion of ethical considerations for technical writers.
- **Chapter 3: Note-taking** provides both the "why" and the "how" of taking notes, including techniques and legal/ethical considerations.
- **Chapter 4: Technical Definitions** explains the "nuts and bolts" of writing effective technical definitions because the ability to define is one of the primary skills needed for most technical writing.
- **Chapter 5: Description of a Mechanism** explains another primary skill needed for technical writing, which is being able to describe mechanisms precisely, accurately, and at a level the audience can understand.
- **Chapter 6: Description of a Process** explains how to describe processes, that is, third-person descriptions of events that do not directly involve the reader.

Section II: Technical Documents

- **Chapter 7: Instructions and Manuals** explains how to write instructions, or second-person descriptions for human involvement that provide specific directions so that the reader can perform a task or series of tasks.
- **Chapter 8: Proposals** explains the three necessary things that all proposals must include, provides multiple examples of informal proposals, and parses the "why" and "how" for each section.
- **Chapter 9: Progress Reports** builds on the proposals in Chapter 8, explaining the necessary elements of a progress report and showing how to construct an effective progress report.
- **Chapter 10: Feasibility and Recommendation Reports** explains how to develop objective documents that identify and evaluate solutions to problems and explains the difference between a feasibility and a recommendation report.
- **Chapter 11: Laboratory and Project Reports** explains the difference between and the purpose of laboratory and project reports, as well as their various elements.
- **Chapter 12: Research Reports** explains the focused, objective nature of research reports, the general structure, and the wide variety of content based on audience need.
- **Chapter 13: A3 Reports** shares the history of the document, templates and examples, and an understanding of the usefulness in business.
- **Chapter 14: Abstracts and Summaries** explains the purpose of abstracts and summaries and provides examples of four different kinds of summations with a focus on both academic and business situations.

Section III: Other Useful Stuff

- **Chapter 15: Style and Mechanics** highlights common issues with basic building blocks, including how style is related to mechanics (spelling, punctuation, and grammar).
- **Chapter 16: Documentation** shows examples of how to cite many different kinds of online, print, and in-person sources.
- **Chapter 17: Visuals** includes basic guidelines for when to use what kinds of visuals, as well as tips for constructing them as accurately and effectively as possible.
- **Chapter 18: Presentations** provides in-person and online presentation tips, including an example set of slides for an update presentation.
- **Chapter 19: Business Communication** provides some general guidelines, outlines, and examples for business communication.
- **Chapter 20: Communication with Future Employers** details six specific kinds of job-seeking communication, including both written (e.g., resumes) and oral (e.g., interviews).
- **Chapter 21: Team Writing** addresses important considerations for accomplishing a collectively written document, both as a student and as a professional.

We believe that our revisions will be useful to students and instructors who choose to use this book. However, we acknowledge (and appreciate!) that language and communication are always evolving and what is considered acceptable today might be adapted by tomorrow. We have done our best to capture generally accepted formulations and long-lived rules.

Being an effective technical writer continues to increase in importance. In fact, we have heard from some employers that they would rather hire good writers with average technical skills (because they can teach the technical skills themselves), than hire someone with high technical skills who communicates poorly. We would be delighted if our textbook could help students gain the communication skills that employers want.

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Department of Industrial and Manufacturing Systems Engineering
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About the Authors

Leo Finkelstein, Jr., received a bachelor's degree from the University of North Carolina at Chapel Hill in 1968; a master's from the University of Tennessee at Knoxville in 1969; and a Ph.D. from Rensselaer Polytechnic Institute at Troy, New York, in 1978. He was Lecturer and Director of Technical Communication for the College of Engineering and Computer Science, Wright State University, Dayton, Ohio. He directed the technical writing program at the U.S. Air Force Academy while also serving as adjunct faculty for the University of Colorado at Colorado Springs. He wrote, produced, and directed technical films in Southern California and commanded a combat-documentation, photographic unit in Southeast Asia during the Vietnam War, flying combat missions as an aerial photographer. In addition, his military service included experience in both space and logistics systems. He held FCC commercial and amateur radio licenses, had a black belt in tae kwon do, and was an avid user of all types of gadgets.

Jeanine (Jenny) Elise Aune is a Teaching Professor and the Director of ISUComm Advanced Communication (AdvComm) program at Iowa State University (ISU). She has an M.A. and Ph.D. from the University of Wisconsin-Madison and has been teaching writing in the Department of English at Iowa State University since 1999. She was the Coordinator of ISU's Learning Community (LC) English links for 11 years. Her responsibilities included helping linked discipline faculty communicate their expectations for students' communication skills to English instructors, and helping English instructors re-design writing classes to help students develop those skills. The number of LC-linked English sections more than doubled during her tenure. She has been directing or co-directing the Advanced Communication program since 2011. She has worked with stakeholders across campus to build standardized curricula for the program's four courses—business communication, proposal and report writing, biological communication, and technical communication—in both face-to-face and online mediums. These four courses are taught by ~40 instructors in ~200 sections and enroll ~4,800 students every academic year. Most recently, the online business communication course earned QM certification[1] at 97%, and the online technical communication course earned QM certification at 98%.

Jeanine Elise Aune

Leslie A. Potter is a Teaching Professor in the Industrial and Manufacturing Systems Engineering (IMSE) department at Iowa State University (ISU). She has a B.S. in industrial engineering from ISU and an M.S. in industrial engineering with an emphasis in manufacturing from The Pennsylvania State University. She worked as an engineer and supervisor for John Deere for seven years before joining IMSE at ISU, where she has taught undergraduate courses across the curriculum for the past 20+ years, ranging from freshman problem-solving and programming to capstone design. As part of her research at ISU, she co-developed a professional communications course within the industrial engineering curriculum. From those efforts, she has developed and incorporated substantial writing and speaking curricula that are used by many of her peers. She was the co-founder in 2013 as well as the co-chair for the IMSE Undergraduate Research program for seven years, supporting hundreds of students with writing and presenting their research. She regularly requires writing and presentation assignments in her engineering courses, and has collaborated with faculty in the Iowa State University Department of English since 2007.

Leslie A. Potter

[1]https://www.qualitymatters.org/reviews-certifications

Email to a Technical Writer

Technical writers must consider their audience, purpose, and situational context. The following is an example of an email that the authors of this textbook could send to potential users of this textbook.

From: JennyandLeslie@Finkelstein
Sent: Monday, June 14, 2021 1:32 PM
To: Student
Subject: Email to a Technical Writer

Dear Student,

Your decision to study a technical field is most admirable, and we applaud your fortitude and motivation in dedicating your time and effort to learning a subject that has the power to change the world. However, your knowledge and expertise can only be understood by others if you can communicate your ideas, thoughts, findings, recommendations, and preferences with both technical and non-technical audiences. It is there, at the point of communicating ideas as non-abstractly as possible, that we can support you. We hope that after reading a few pages of our not-typically-super-serious textbook, having a conversation or two with others about it, and allowing yourself to enjoy the relatively formulaic processes of technical writing, you will come to appreciate the power of audience, purpose, and context. With this note, we offer two thoughts.

First: Writing in the real world is very different from the writing you have likely done to date.

You will need to change your mindset. No longer will you write for an audience of one. No longer will you write to a teacher to show them how much you know or have learned. We must now ask you to put aside the unintended mindset you might have of, "I will write what my instructor wants." To become an effective technical writer, you must anticipate who your audience will be, recognizing that you might have multiple audience types, and why you will write for each of them. Do you wish to inform, persuade, or simply create goodwill? Ask yourself: what outcome do I need? And how can I make it happen? Then write with this at the front of your mind.

Second: While technical writing is a serious endeavor, one (and by that we include you) should not take oneself too seriously.

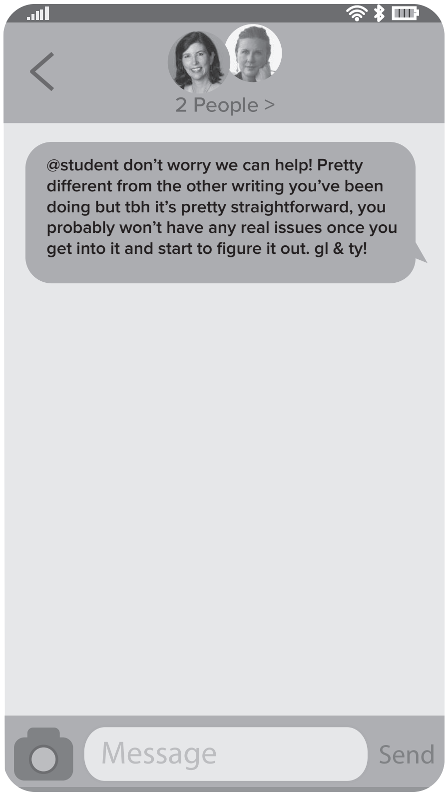
Accomplishment is one of life's greatest rewards. Perhaps you have already been successful in calculus, soccer, robotics, speech, origami, research, baking, dance, or gaming. You sit in our classroom—we know how amazing you are! And we know that you can be an accomplished technical writer, too. As with your other successes, it requires only a willingness to improve through practice and reflection, but this is ever-so-much-more enjoyable with a quick laugh and permission to enjoy the sometimes painfully iterative improvement process. Somewhat related, we thank you for indulging our, shall we say "quirky," sense of humor (some will say "bad"—we happily accept that). In short, we say lighten up! Enjoy the ride. And the write!

We thank you for the confidence you will place in our ability to communicate about communication, knowing full well that the initial decision was not your own, but your instructor's.

Yours very truly,
Jenny and Leslie

IM to a Technical Writer

The same information can be presented in different ways. For example, the authors of this textbook could also choose to send an IM to potential users of this textbook. The basic idea is the same, but much detail has been removed and the language is much more informal.



Acknowledgements

Acknowledging those who have helped us with the 4th edition of *Technical Writing for Engineers and Scientists* is a daunting task. While our names are listed as authors, they are only listed there due to the opportunity provided to us by McGraw Hill and the support of our families, friends, and colleagues.

Thank you to McGraw Hill for accepting our proposal for revising the textbook and giving us the opportunity to teach on a broader scale than our classrooms allow. Thank you to Shannon O'Donnell for the preliminary conversation a couple of years back about the potential for a revision of the textbook and supporting our authorship, Theresa Collins for getting us started on the process, and Erin Kamm for guiding us through to the end. In addition, thank you to Beth Bettcher for supporting our endeavor; Beth Cray for helping gain permissions; and Maria McGreal, Dheeraj Kumar, and team for getting our image plans to paper. There are so many elements, tiny details, and untold hours of work that go into creating a published textbook from a text document; hopefully, we have not missed anyone involved in that monumental task.

Thank you to our families for both their general support and specific feedback. They graciously tolerated our late nights uploading and re-uploading documents, working on laptops while watching family movies, and phone calls while out walking our dogs. They patiently listened to ideas for, and drafts of, our dog analogies. They provided honest feedback and even managed to find our excited texting back-and-forth over dinners amusing. Both Siggi (Jenny's husband) and John (Leslie's husband) provided helpful technical confirmation in their own areas of expertise, as well as IT support when needed. Thank you to Kathy Baker (Leslie's mom) for proofreading our drafts with an eagle eye. We also especially thank our oldest children, Sunneva Sigurðsdóttir (Jenny's) and Jack (Leslie's), for reviewing and commenting on sections of the textbook which, frankly, required the expertise of college-age students to appropriately address the advances of online communication platforms made since the last edition (you know, like, 'cuz we're "boomers" . . .).

Several colleagues helped inform our revision of this textbook; thank you to them for providing another level of expertise and for always being willing to help when asked: Sarah Ryan, Frank Peters, Michael Helwig, Dave Sly, Christopher Proskey, Valerie Boelman, and Joseph Schneider. Thank you, also, to our students who provided their actual resumes for us to adapt and use in examples and exercises.

And, last but not least, we thank all the students with whom we have had the privilege to work over the years. Each and every exchange was an opportunity for us to learn how to explain concepts, provide feedback, and help motivate to improve their communication skills. We may not have always been successful, especially in those early years, but we were eager to improve ourselves, thought about how we could have done better, and to this day, still continue to learn.

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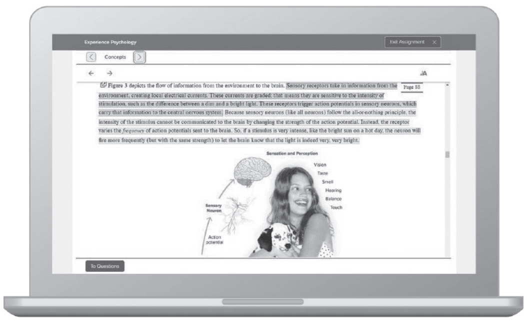

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CHAPTER ONE

01

00001

Introduction

Technical writing is a fundamental skill for virtually everyone working in science and engineering—and that includes a broader range of people than just scientists and engineers. Most science and engineering activities require technical communication, whether in written, electronic, or some other form. Research, development, finance, manufacturing, and a host of technical commercial services rely on precise communication to relay complex information to a wide range of audiences for many purposes. Technical writing is the means by which such communication is produced.

What Is Technical Writing?

To define what technical writing is, it might be useful to first clarify what it is not. Technical writing is not what one does out in the meadow under an elm tree; that is creative writing (unless, of course, you are a meadow habitats specialist!). Creative writing, while challenging, is an artistic activity. It flourishes in a world of nuance, interpretation, and subtlety. Technical writing, however, is the opposite of artistic. Technical writing is writing with precision and eliminating the possibility of (mis)interpretation. Technical writing is not what most people commonly do for fun or relaxation.

Imagine, if you will, a learned, sophisticated person sipping fine wine, eating imported cheese, and watching the sunset. This person sounds like a creative writer who, with paper and pen in hand, might reflect on the nature of time as the last rays of sunlight slowly disappear over the horizon. Our creative writer might even develop a definition of **time** that goes something like this:

> **Time** is a river flowing from nowhere[1] through which everything and everyone move forward to meet their fate.

A creative, sensitive person sees an inspiring sunset, is moved to words, and writes the "river from nowhere" definition. Obviously, this approach is metaphorical. But what if someone inspired by that magnificent sunset were to write the following definition of **time**?

> **Time** is a convention of measurement based on the microwave spectral line emitted by cesium atoms with an atomic weight of 133 and an integral frequency of 9,192,631,770 hertz.

Perhaps not! It would certainly take a unique individual to be emotionally moved by a sunset and then come up with microwave spectral lines and cesium 133. The second definition of time is technical. It is designed to be objective, direct, and precise. Consequently, it lacks the emotional impact of the first definition because, as technical writing, it avoids the use of rich metaphors and figures of speech, substituting instead precise, empirical facts.

The difference between the two definitions shows the fundamental distinction between technical writing and creative writing (and all the other kinds of writing that fall in between). Technical writing is precise, objective, direct, and clearly defined.

Reducing Abstraction

In linguistic terms, technical writing is writing that displays a relatively low level of abstraction. To clarify the concept of abstraction, consider the funnel of abstraction in Figure 1.1. Here, various levels of abstraction are being used to refer to the specific merle pigmentation of your friend's new puppy, Sparky-the-Merle. Imagine that you are highly allergic and entirely unfamiliar with anything dog-related, but your friend is extremely excited about their puppy being a merle. They explain that merle is a **dog genetic mutation.** However, that does little to help you understand their excitement—"dog genetic mutation" is a high-level

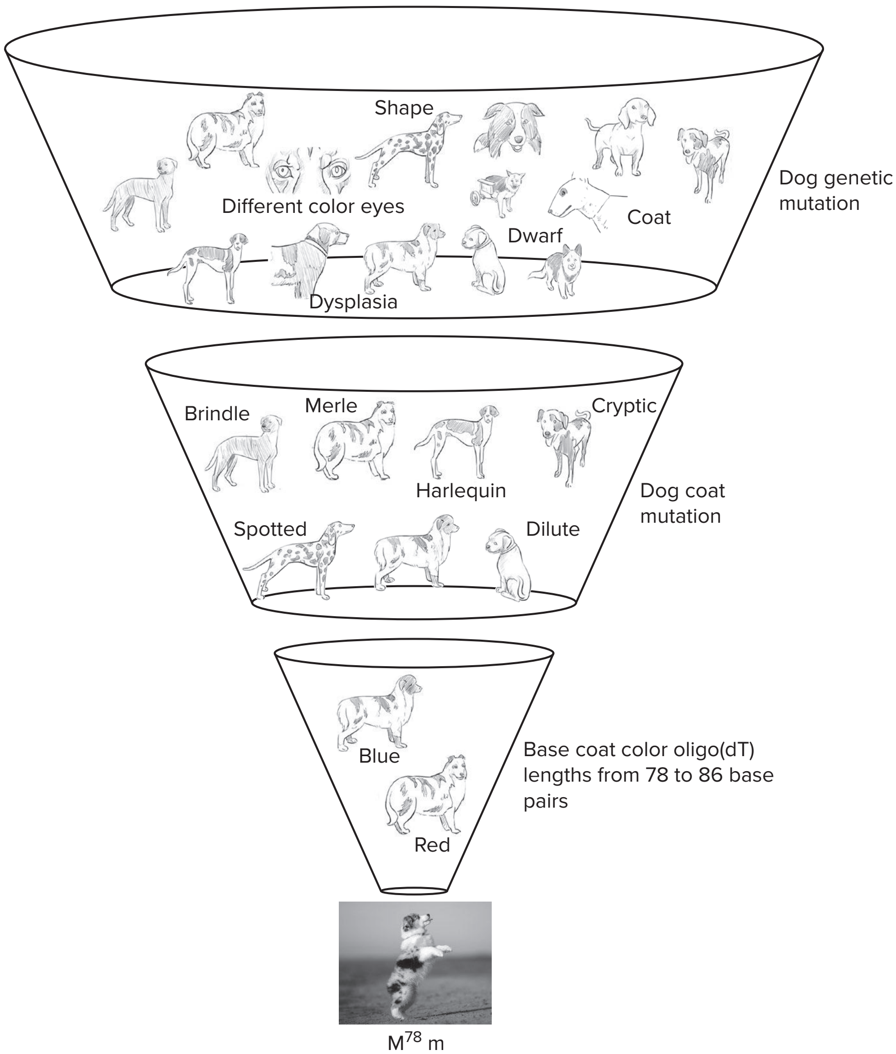

Figure 1.1
A funnel of abstraction. The high level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.
otsphoto/Shutterstock

abstraction and could be anything from all kinds of dog coat variations to unmatched eye color to hip dysplasia (as seen at the top of the funnel in Figure 1.1).

When you ask for more clarity, your friend explains that **merle** is a term used to describe a dog's splotchy coat, a type of **dog coat mutation,** which could mean any type of coat coloring with irregularly shaped spots. This description allows you to envision what merle means, and could include a dog with a coat that fits in many different categories of merle, such as cryptic, dilute, harlequin, or, in this case, standard. As we filter the level of abstraction, by adding some objective precision in our words, the reference becomes increasingly precise.

Imagine how lucky the happenstance that you and your friend are both geneticists, and you both quickly get to an even more filtered and precise description. Your friend refers to the pigmentation of a standard merle dog, which is described as those with **base coat color oligo(dT) lengths from 78 to 86 base pairs.** You can specifically describe the genetics of a dog's coat type by using words alone.

The lowest level of abstraction would be Sparky's actual DNA. However, we normally do not paste genetic material into technical documents; so to be precise, we must substitute something else, such as a photograph of his coat and a standard description of his genetics, which in this case is $M^{78}m$. That is why a photograph of Sparky the merle-coated dog has been placed at the narrow end of the abstraction funnel—because that is the most concrete way available to refer to it in a document.

Let's look at another example. Imagine that you need an explanation of a 33K, one-watt carbon resistor and you read that it's an **electrical device.** Not helpful. "An **electrical device**" could be anything that is electrical, from a fluorescent light, to a computer keyboard, to your phone's SIM card (see Figure 1.2). Of course, mixed in with all those electrical devices is the electrical device also referred to as a 33-kilohm, 1-watt carbon resistor, but it would be difficult to hone in on it with all the other noise of **electrical devices.**

You read on that it is a **circuit component.** That's certainly more precise, but it could mean any electrical device in the circuit, such as a capacitor, inductor, diode, or, in this case, a resistor. As the explanation becomes more precise, it includes the word **resistor,** which means any circuit component that impedes the flow of current.

The next most concrete way of describing the resistor by using words alone is to precisely label it—as we do when we give our newborn children distinctive names and government taxpayer identification numbers. In this case, the resistor is labeled a **33-kilohm 1-watt carbon resistor.**

As in the case of Sparky, the lowest level of abstraction would be the resistor itself, which we cannot paste into a technical document. We substitute the resistor with a photograph, which is why a photograph of the resistor has been placed at the bottom of the funnel. This precision, of course, assumes that the audience knows what a resistor looks like. If not, the photograph can be pretty abstract, and for that matter, so can the actual resistor.

In fact, when we write technical documents for our audience, we need to carefully consider our audience's familiarity with our topic and carefully calibrate the level of abstraction to the lowest level that will help them understand given their current knowledge of the topic. Technical writing is **not** about how well the writer understands the topic; technical writing is about **how well the writer can explain the topic to someone else.**

As one moves down the funnel of abstraction, the symbols become more precise and less vague. In effect, this reduced abstraction gives the reader less freedom to interpret meaning as they want and more like the writer intended. In creative writing, this lack of flexibility is probably not good; in technical writing, such lack of flexibility is good. The goal of technical writing is to eliminate abstraction. Simply speaking, successful technical writing restricts the reader's freedom of interpretation so that only one meaning can be concluded—the meaning intended by the writer.

What sets technical writing apart, then, is its precision. How it achieves this precision is, in fact, the art and craft of technical writing—an activity that involves definition and description; data and analysis; photographs, diagrams, and charts; and often specialized language. The goal of technical writing, then, is **not** to be creative, or interesting, or to employ rich imagery or powerful metaphors. The goal of technical writing,

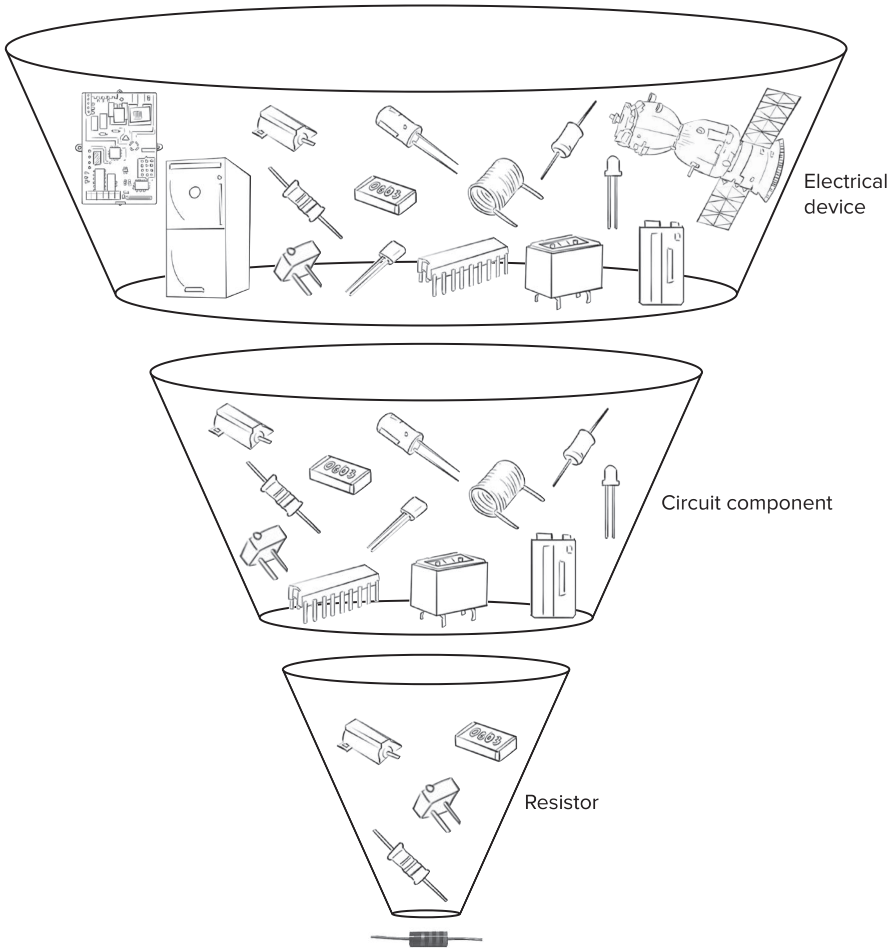

Figure 1.2
A funnel of abstraction. The high level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.

first and foremost, is to communicate complex information clearly and precisely for the audience and the purpose at hand. Clarity and precision are the overriding goals for any technical writer, and understanding the audience, purpose, and context is the primary consideration for achieving those goals.

Audience, Purpose, and Context

The measure of how well a technical writer has written something depends on three things:

1. how well the **reader** understands, precisely, the writer's intended meaning,
2. how well that understanding fulfills the intended **purpose,** and
3. how well the communication fits into the broader situational **context.**

Consequently, technical writing must be geared directly to its audience, purpose, and context. Remember, there is always a specific requirement for technical writing: a scientist must write a grant proposal, a programmer must document software prior to distribution, a lab supervisor must justify new equipment with a feasibility report, and an industrial engineer must convince a business manager to change work practices for the long-term health of employees.

Relating audience, purpose, and context requires that the writer consider the potential reader's knowledge, skill level, and specialization. The writer must anticipate fully responding to the needs of the reader in terms of the requirements of the situation. Let us take a closer look at each of the three separate yet interrelated components.

Audience

Technical writing is only effective if the **audience** understands the communication. To ensure that our writing is understood by the audience, we must contemplate and often even research our audience. Some things to consider include the following questions.

- How much does my audience know about the topic?
- How much background information about the topic might my audience need to understand my communication?
- How motivated is my audience to read my communication?
- What level of detail does my audience need to understand or to accept my communication?
- What are my audience's primary, secondary, and tertiary concerns?
- What concepts in my message would be better and more quickly understood with supporting visuals?
- How much documentation will my audience want and/or require?
- What format does my audience expect?
- How much time will my audience have to read my communication?
- Where will my audience use my communication and how does that affect how I should structure and format my communication?
- Will my audience be resistant to my communication? (the idea, message, topic, etc.)
- Will my audience be resistant to the communication coming from me personally?

Consider the industrial engineer who must convince a business manager to change work practices for the long-term health of employees. How will they answer these questions about the audience? For example, if there have been employees missing work because of carpal tunnel injuries, the business manager might be very receptive to a plan for reducing absenteeism. But if the same manager is struggling with immediate warranty costs caused by quality issues in an assembly area, long-term concerns like carpal tunnel may seem less urgent in the moment.

Purpose

Technical writing must fit with the **purpose** of communication, which is why you are communicating in the first place. There are several things to consider, starting with what we want our audience to do after they have read our communication. Do we want them to

- know more facts?
- better understand how something works?
- know the status of a large project?

- provide money for a project?
- assemble something?
- accept a recommendation?
- change their position on an issue?
- make a decision?
- some combination of things?

Once we have determined what we want or need our audience to do after they have read our communication, we must determine how to get our desired result.

- Do I need to inform? If so, to what level of comprehension?
- Do I need to persuade? That is, do I need to change someone's position on a topic, their belief about something, or their behavior or actions? If so, is there information that I should emphasize or highlight?
- Do I need to build goodwill? Sometimes this is a necessary requirement by itself, but other times we incorporate this into our communication which informs and/or persuades.

Again, consider the industrial engineer who must convince a business manager to change work practices for the long-term health of employees. How will they answer these questions about purpose? For example, they might need to talk with machine operators to get more information, but that means overcoming a possible mistrust of management. Once the industrial engineer has the necessary facts, they must think about how to request money from their manager. Is goodwill already established? Can they move to considering how to persuade?

Context

In addition to relating to a specific audience and for a specific purpose, technical writing must fit within the broader situational context. Unfortunately, much of the broader context is outside our control as communicators, yet we must account for it nonetheless. Context includes things like history, language, geography, politics, culture, economics—basically anything in the world that affects how our communication is received by our audience. It could be something as specific as personality: maybe someone in management doesn't really like you, and you don't know why, but you know that they immediately respond negatively to anything you suggest. It could be something like company culture: perhaps you tend to write informally and use contractions, slang, and emoticons for emphasis ☺, but the company culture is formal. It could be something historical, like a chemical accident that killed an employee over 20 years ago—it is not a topic that is actively discussed, but it lingers deep in peoples' minds. Communication does not exist in a vacuum.

Let us go back to the industrial engineer. What contextual factors might affect the engineer's ability to convince their business manager to change work practices? Perhaps a recent OSHA law was passed that mandates regulations be implemented by a certain date or a company will be fined. Maybe a competitor in the industry had a lost-time accident occur and suddenly other companies take notice. What if there is a tight job market and employees care about safety and use it to decide where to go to work?

In technical communication, the audience and purpose are almost always well defined in advance—often by the supervisor, customer, or teacher. This is because technical writing is usually commissioned by someone else for a specific purpose and audience. Normally, technical communication aims to share objective information with an interested, educated audience. Technical communication is simply communication on technical subjects that shares information in a precise way.

However, the broader contextual situation, or context, is usually outside a technical writer's control. Remember our industrial engineer who wants to convince a business manager to change work practices for the long-term health of employees? Well, the engineer did their due diligence on researching the frequency and severity of injuries, the biomechanics of how the injuries happen, and the audience's concerns. They wrote a precise, meticulous, eloquent feasibility report. However, behind closed doors, upper management had discussed the closing down of that assembly line in the next three to four years. Despite the impressive quality of the engineer's report, it was shelved due to a factor outside the engineer's control. The engineer didn't have all of the necessary context needed to write an informed proposal.

Genre Decisions

Technical writing must work within the parameters of genre expectations in addition to relating specifically to the audience, purpose, and context. Genres are categories of things that have a similar subject matter, form, and style. You might not have heard of the term **genre** before, but it's a concept that you likely use every day.

- You are going for a run, so you listen to high-energy rock songs
- You are going for a three-hour drive, so you play podcasts on scientific discoveries
- You have had a bad day at work, so you kick back with an action movie

We all choose categories such as these, called **genres**, because we know what they are (generally speaking) going to be about, how they will be organized, and what they will sound, read, and feel like.

When we think about categories, or genres, that are understood the same way by people from all over the world, we can draw an analogy to dogs. Dogs, just like communication, have been around for thousands of years, and both have evolved with the right breed/tool identified for specific needs and jobs. When you need a mouser, you look for a terrier. When you need to herd livestock, you get a herding dog. When you need to retrieve water fowl, you choose a sporting dog. From the Chihuahua to the Collie to the Great Dane, you recognize all of them as dogs, but you also know that they do different things and have different purposes (see Figure 1.3). The same is true in communication. When you need to convey information to a certain kind of audience in a particular way, you choose the appropriate formulation. When you need to get funding for a project, you write a proposal. When you need to explain how to put that new basketball hoop together, you write instructions. When you need to get an internship or a job, you write a resumé. Each of these technical writing genres has a specific purpose or function.

However, once you've chosen which technical writing genre is most appropriate for your audience, purpose, and context, you still have many decisions to make. You must appropriately adjust the amount of content, level of detail, structure, format, tone, style, and length of each element of that genre in your particular document. Consider the farmer who needs a herding dog—what other factors do they need to consider? Well, to start, what type of animal will the dog be herding? Under what climate conditions? Will it also be a family dog? How experienced is the farmer with training dogs? How protective should the dog be?

Figure 1.3
Much like different dog breeds have specialized purposes (to hunt, to herd, to protect, etc.), communication genres have specialized purposes (to inform, to persuade, to build goodwill, etc.).
belchonock/123RF

Does it matter how loud the dog is? How much it sheds? How many years does the dog typically work? There may be other factors to consider, too, but the farmer shouldn't randomly choose one of the approximately 70 breeds in the herding group. The farmer should carefully consider as many contextual factors as they can and make strategic determinations. Similarly, as a technical writer, to communicate effectively, you will also need to make decisions about elements of the genre you have chosen.

Characteristics of Technical Communication

So, finally, to answer the question posed at the beginning of this chapter—What is Technical Writing?—technical writing can be characterized as follows:

- **Technical communication is non-abstract and precise.** The goal of technical communication is to eliminate abstraction and communicate complex information clearly and precisely to the audience at hand.
- **Technical communication considers audience, purpose, and context.** Technical communication must not only relate specifically to the audience who will be using the communication, but also relate specifically to the purpose and context at hand.
- **Technical writing deals with technical information.** Technical writing is designed to deal with technical subjects.
- **Technical writing relies heavily on visuals.** As shown in the abstraction funnel of Figures 1.1 and 1.2, a photograph or diagram may be the least abstract, and therefore the most precise, way to communicate something to an audience. Visuals—whether they are equations, photographs, tables, graphs, or drawings—are powerful tools for presenting a large amount of information effectively and efficiently. However, they almost always require interpretation or explanation in the text of the report, especially when the audience may not be familiar with their content.
- **Technical writing uses numerical data to precisely describe quantity and direction.** In many cases, mathematical equations and formulas, experimental results, and financial analyses provide the real substance of a technical report.
- **Technical writing is accurate and well documented.** Generalized, unsupported assertions have no place in technical writing. Conclusions, recommendations, and judgments are always based on clearly presented evidence or established expertise, and technical writing is always technically correct.
- **Technical writing is stylistically and mechanically correct.** This kind of correctness is far more than simply a matter of personal and corporate pride. Style and mechanics (spelling, punctuation, and grammar) often go to the heart of the author's credibility. In other words, right or wrong, fair or unfair, if you write shoddily, the reader may well perceive you as careless and your organization as a collection of people who do not care about details. Whether or not this perception is true is irrelevant. Where technical writing is concerned, perception is often more important than reality.

Notes

1. The idea of a "river from nowhere" came from the Steve Winwood song "The Finer Things," which includes a reference to time as a "river rolling into nowhere."

CHAPTER TWO

02
00010

Ethical Considerations

Be the person your dog thinks you are.[1]

Why is ethics the focus of a lead chapter in this book? It is here because of increasing emphasis being placed on ethics in technical writing, as well as in engineering and the sciences as a whole. Technology and the communication of technical ideas represent powerful tools for promoting good or evil in society—and ethics is an important backdrop for all the material provided in the chapters that follow.

Ethical considerations are not separate from the work that engineers and scientists do, but are entwined in it. For example, the American Chemical Society has a standing Ethics Committee; one of their goals is to "increase the members' recognition of ethics in their careers and its impact on society."[2] In another example, many undergraduate engineering programs are accredited through ABET, which requires program achievement of "an ability to recognize ethical and professional situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts."[3] Boston University published "The Ethics of Scientific Research—A Guidebook for Course Development" (Stern and Elliott), which includes the objectives: "a. Keep good notebooks" and "d. Record and report your work accurately" 1997.[4] Companies strive to be named on the list of "World's Most Ethical Companies"[5] and publish their own reports about what ethics means to them (*Ethics in Scientific Research—An Examination of Ethical Principles and Emerging Topics*), Rand Corporation 2019.[6] Ethical considerations are an inescapable part of our broader human context.

Humanity has been struggling for a long time over exactly what constitutes good and evil, and ethical and unethical behavior. It is unlikely that this chapter in a technical writing book will provide the definitive answer that will end this timeless struggle. The hope, however, is that this chapter will provide an awareness of the ethical dimensions of technical writing and some sensitivity to the real issues that exist within these dimensions. Additionally, this chapter will discuss several key areas of ethics in technical writing, including ethical writing, academic and professional dishonesty, copyright infringement, image alteration, and personal online presence.

What Is *Ethics* in Technical Writing?

The first thing we must develop is a working definition of ethics in technical writing. What exactly is ethics for a technical writer? For the purposes of this book, the answer is relatively simple. Collectively, **ethics** is a set of rules and standards for using communication skills and resources with the intention of doing good. Ethical behavior for technical writers, then, involves their moral duty and obligation to apply the power of technical communications to the purpose of doing worthy things. But what do **good** and **worthy** mean?

You probably already know what you think they mean, but not everyone would agree with you. The global technical writing community is composed of many disparate cultures. To some extent, concepts of good and bad, and ethical and unethical, are culturally relative. However, that does not mean we can just ignore the issue or take the viewpoint that our ethics are better than others' ethics. We must approach the matter realistically in the context of the diverse world in which we function. That is what this chapter attempts to do.

Begin by considering the following situation:

> Three technical writers write an irreverent, humorous technical writing book using, as its examples, reports on fictitious technologies based on real theory. Their laboratory report chapter deals with a fictional, high-power transmitting tube called the *SuperXLTube,* their research report chapter describes a quantum computing processor known as the *QuantumCPU,* and their process description chapter includes a figure skating jump called the *QuadFINKEL.* For the sake of argument, assume that the book is a big hit, the royalties pour in, and the authors deposit the checks and live the good life.

Now consider this situation:

> Three technical writers write a professional proposal from a fake corporation. They invite individual investment of funds in developing and marketing the fake company's fictitious product line of high-technology devices. They cite fabricated test data and technical research information demonstrating the efficacy of these products, and they develop altered photographs of the product line. In fact, they develop a professional, full-color brochure complete with tables of data, charts, diagrams, and photographs—and promise huge returns on investment. They mail this brochure to thousands of nursing home residents across the country. When many of these people send the technical writers their life savings, the authors deposit the checks and live the good life.

In both situations, the technical writers have used their knowledge and resources to develop published materials around fictitious technologies. In both situations, they have embellished the capabilities of these technologies with professionally developed photographs, diagrams, charts, and data. In both situations, they make money from the published product.

In ethical terms, how do these situations differ? That is an important question; the difference gets at what makes the behavior of technical writers ethical or unethical. Before we focus on this difference, however, we need to look briefly at the kinds of ethical constructs that are traditionally used in technical writing and the implications of these constructs for technical writers:

- **Technical writers must be accurate in their work.** Either technical writers must be precisely correct at all times, or they are unethical.
- **Technical writers must be honest in their work.** Technical writers who write untruths are unethical.
- **Technical writers must always honor their obligations.** Technical writers who do not produce the documents and other materials they are responsible for producing within the agreed-upon time frame are unethical.
- **Technical writers must not substitute speculation for fact.** Technical writers who do not clearly separate opinion from accepted truth are unethical.
- **Technical writers must not hide truth with ambiguity.** Technical writers who either omit, deemphasize, or misrepresent facts that would be contrary to the theses of their reports are unethical.
- **Technical writers must not use the ideas of others without giving proper credit.** Technical writers who fail to document the sources of all nonoriginal ideas, except for common knowledge, are unethical.
- **Technical writers must not violate copyright laws.** Technical writers who fail to document the use of copyrighted materials when used with permission or under "fair use" are unethical. Additionally, technical writers who use any copyrighted materials without permission when these uses are not covered by fair use are unethical whether the materials are documented or not.
- **Technical writers must not lie with statistics.** Technical writers who manipulate data or graphical representations of data, use inappropriate or improper statistical tests, or employ loaded statistical samples are unethical.
- **Technical writers must not inject personal bias into their reports.** Technical writers who are less than objective in everything they write are unethical.

What a list! These rules are clearly well intentioned, and they provide generally useful guidelines for writers. But there is a real problem with rules like these: they miss the mark where ethics are concerned. For example, being accurate and precise in technical writing is not the real ethical issue here. Intending to **be** good and to **do** good are the real issues. Of course, it is true that in technical writing "being good and doing good" usually mean being accurate and precise, but not always.

In the first situation, as the authors in question, we obviously do not believe that being accurate and precise is necessary to achieving our pedagogical goals for this book. In this case, we believe that just the opposite is true. The freedom to invent obviously fictitious technical writing examples supports the book's educational objectives. We are being inaccurate, but we are not being deceptive. For example, in the process description chapter, an exercise deals with the QuadFINKEL figure skating jump, an athletic achievement so technically demanding that anyone landing it always receives the gold medal. Do you suppose anyone really believes that a QuadFINKEL exists? In fact, most readers probably have serious doubts as to whether the authors can even stand up on figure skates—and these doubts would be well-founded.

We would be deceptive, and therefore unethical, only if we asked that you, the reader, truly believe these examples. Our clear intent is not to mislead; rather, it is to use the fictitious and humorous examples to help students learn—and we believe learning is still considered a "good" thing to do. However, we wouldn't do well with that list of rules, would we? These rules do not get at ethics; rather, they get at behavior that is often correlated with ethics.

In the second example, the rogues trying to rip off nursing home residents are obviously not the kind of people you would want managing your money. They clearly intend to use their technical communications skills and resources to deceive. To make matters worse, they are also targeting a vulnerable group of people for this scam—the elderly, who are least able to afford this kind of attack. So unlike in the first situation, the tricksters are falsifying information not to do good, but to do bad. That is why the technical writers in the second example are absolutely unethical (not to mention being guilty of violating numerous local, state, and federal laws).

Figure 2.1 presents a model for ethics in technical writing. Clearly, it does not represent the big breakthrough in the philosophy of ethics for which humanity has been waiting so long. But it does help make the point and clarify the issue. The rules may contribute to ethics and our understanding of what is involved in ethics, but they are not what constitute ethics.

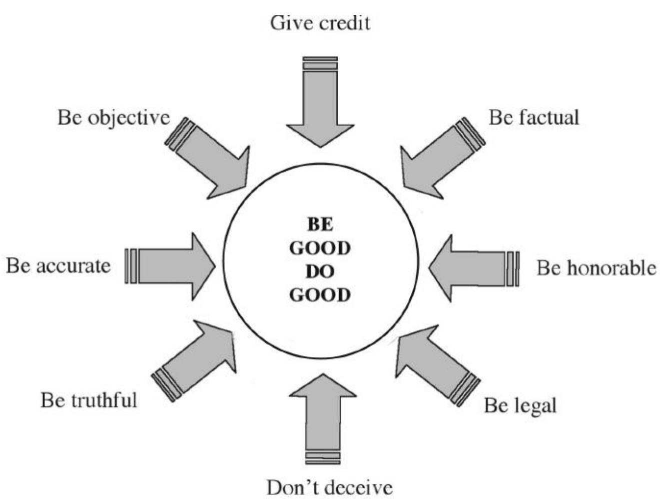

Figure 2.1
Ethics model for technical writing.

Finally, ethics in technical writing, to some extent, must be relative to societal values. If, on the one hand, you use technical communications to deceive your reader with the goal of doing bad things (defining bad by the standards of your society and perhaps by the standards of civilization as a whole), you are unethical. If, on the other hand, you use technical communications with the intention of doing good things (defining good in the same manner), you are ethical.

Academic and Professional Dishonesty and Copyright Infringement

When you submit work in a class—whether it be a paper, test, quiz, homework assignment, or something else—you are saying that the work you put into that submission was your own and that you followed the instructor's requirements. Using someone else's work without acknowledging the source constitutes **academic dishonesty** in a school setting and **professional dishonesty** in industry. Academic and professional dishonesty encompasses multiple types of stealing, although the types might have different labels:

- **Using dishonestly obtained information.** "Dishonestly obtained" could be anything from wholesale copying of someone else's work (from a friend, a student sitting in the chair next to you, an online study site, a co-worker, or a competitor); buying work from a student, a writer contracted to write for you, or from an online source; working with others on an individual project; or even sneaking a disallowed peak at your own notes during an exam. If you use these methods to complete your own work, you are being dishonest.
- **Helping others get dishonestly obtained information.** Even if you are not the student or employee actually submitting the work, you can still be academically or professionally dishonest. This type of dishonesty includes giving or selling your own work—whether it be homework, exam questions, data, or papers—and sharing course materials developed by your instructors with an online study site. If you are helping others to be dishonest, you are being dishonest.
- **Failing to participate.** If you fail to participate in an assigned group project and make the other group members do all of the work, you are being dishonest.
- **Making up data and evidence.** If you fabricate data instead of conducting an experiment and recording the results, or fabricate evidence instead of researching a topic, you are being dishonest.
- **Gaining academic and professional advantage dishonestly.** If you destroy the work of others so they cannot complete the assignment (or complete it on time) or if you try to offer money, gifts, or services to others in order to gain an academic or professional advantage for yourself, you are being dishonest.
- **Stealing an idea or an expression of that idea (plagiarism).** If you steal a person's idea, or their expression of an idea—whether in words, images, sounds, photographs, etc.—and then represent it as your own, you are a plagiarist. (For stealing the expression of the idea, you could also be guilty of a copyright violation.)

All of the acts listed above aim to deceive others into thinking the work submitted by the student or professional is original work.

The problem of academic and professional dishonesty has increased dramatically because of the internet and technology. Many years ago, people who wanted to get information dishonestly or copy the works of others for an assignment had to go to the library and look through a limited collection of materials likely known to their teachers or supervisors. Students or professionals might even have had to use a copy machine (sometimes involving many dollars' worth of quarters) or endure the drudgery of transcribing the material manually. They would have to physically share papers or gain physical access to data or exams (perhaps even breaking into an office or file cabinet of an instructor or coworker!). When someone copied another's work, the act itself involved time and effort, and included the ever-present risk of detection. In many cases, plagiarism actually cost more in time and money than it was worth.

Those days are over. Now the internet and its multiple search engines have extraordinary search capabilities that provide people with quick access to literally billions of sources and documents. With a press of a touch pad, the drag of a cursor, and a few simple key strokes, any image, graph, or line of text can be copied and pasted into a plagiarist's document in privacy. For some, copying material from the internet may seem less wrong than copying pages from a book. Some see information available on the internet as public property that is theirs for the taking—not unlike old furniture a neighbor has put out at the curb on trash collection day. The ephemeral nature of files on the internet, the lack of material tangibility, the speed and impunity with which this material can be located and reproduced, and the ease with which it can be used in private are all factors that contribute to the perception that somehow stealing a file from the internet is less wrong than copying from a hardcover journal or book.

The risk of detection for the copy-and-paste type of plagiarism is increasing as academic institutions employ the use of plagiarism detection software. Such software compares submissions against previously submitted work and content available on the internet. It produces a report showing which portions, if any, of the submissions are copied and the sources from where that material was copied. This does not mean the end of academic dishonesty and plagiarism, however. Plagiarists and cheaters can now hire essay writing services to write custom work that students can pretend is their own and submit for classes. The increase in essay writing service sites for students is so plentiful that there are now reviews to help students choose which service to hire.[7] It shouldn't need to be said, but if you hire one of these sites to complete work for you, you are a plagiarist. By definition, this makes you dishonest.

For some struggling with the ethics of plagiarism, the fact that "everyone" seems to be doing it somehow makes doing it more acceptable. That is a logical fallacy, more specifically a bandwagon fallacy,[8] and just because it seems like "everyone" is doing an unethical thing, does not magically transform an unethical action into an ethical action. It is still wrong. Fortunately, most students and professionals are not ethically challenged.

Are there implications beyond the classroom? Yes. More and more employers, institutions, and external agencies are requesting student files prior to offering full-time employment because they only want to hire ethical employees. For example, law schools and government or military agencies will often request that students sign a waiver or release of information to ask for information on a student's disciplinary record at their university, even if that information wasn't provided on an official transcript.

Academic and professional dishonesty is unethical and has serious consequences, but it is not illegal. Copyright infringement **is** illegal. **Copyright** is a type of intellectual property protection that provides federal and international protection for the expression of ideas, but not for the ideas themselves. When someone disregards, or infringes on, a creator's copyrights, that someone is guilty of **copyright infringement,** an act that has legal consequences. Copyright laws protect "original works of authorship"[9] by giving their creators a "set of rights to protect their work" and "to protect the creator from OTHERS who might be doing the same without permission, attribution, or payment"[10]:

- The right to reproduce the work in copies
- The right to create derivatives of the work
- The right to perform the work publicly and digitally
- The right to display the work publicly
- The right to the work digitally
- The right to let others exercise these rights

Before electronic publishing and the internet, the expression of ideas basically took the form of images printed on a page, whether those images were text, photographs, or illustrations. In other words, the ideas were encoded into traditional media and were protected in that form. In many cases, it was impossible to separate an idea from its expression (as with, say, a photograph)—and publishers exercised substantial control over what the expression looked like, what form it took, and where and how it was distributed.

In effect, the limitations of the medium provided reasonably good protection for the ideas it carried. You could copy or duplicate a printed page, an illustration, or a photograph—but you had to have an original or good-quality copy to do so. After one or two generations of "analog" duplication, the copy would become so degraded that it would not be worth having. And, of course, copying usually was not free. These inherent limitations helped to reduce the widespread pirating of copyrighted material that is much more common today.

Now when an electronic document is placed on a web server, it is likely accessible to anyone anywhere in the world. Viewing it requires that it be downloaded and copied to the viewing computer's memory—and probably, as well, to the browser's cache on the hard drive of that computer. Anyone viewing a web document has no need to make a local copy because, given the nature of the technology and the way web browsers work, that copy already exists. One needs only to specifically name and save it as a file to permanently retain an exact duplicate of the original. The hypertext, graphics, and sound files are just as good, quality-wise, as those on the original server from which they were taken.

Consequently, copyrighted material on the web is often available to anyone who wants it, and copies can spread rapidly with little or no degradation and virtually no control. It is imperative that you consider any technical report you place on a web server to be public material in a global information infrastructure; expect that people will look at, copy, and use it freely. Of course, you could encrypt the documents, password-protect them, or even place the materials on a secure server. But these inherent limitations to access may undermine the purpose of having the materials online in the first place. Moreover, no protection scheme is totally hackproof.

Other possibilities for protecting electronic documents exist, including digital watermarks and various licensing schemes. But when all is said and done, the internet is still a global community—and the rules and values you embrace may not apply everywhere. For copyright protections to be effective, they must be international and enforced for all users who have access to the information. That may be where we are headed, but we are not there yet.[11] Consider carefully what you put online and why you are putting it there, before you actually do so.

What does all of this mean for you as a user of copyrighted material? It means that you cannot share movies or songs—or even large portions of them—without the copyright owner's permission. Copyright is the reason that fans of the breakthrough TV show *Northern Exposure* cannot view it on a streaming platform; "the show's creators didn't want to compromise by replacing the music used at the time with generic music (like you'll hear in streaming versions of *Alias* or *Beverly Hills 90210,* for example)," but using the original music means paying fees to the songs' copyright holders and "no one wants to pay artists for rights to use their songs in the streaming version."[12] Copyright means that you cannot scan, duplicate, copy, and then use an image (photograph, drawing, etc.) without the copyright holder's permission or without citing it, and copyright is why you can't use a corporate logo in your own works without the company's permission. Copyright is why you cannot make copies of or share games, apps, or software. Copyright is also the reason that instructors cannot save you money by just posting copies of course reading materials, whether original electronic publications or scans of original print publication, for you to access for free.

Copyright laws protect the creator's rights and the consequences of infringing on those rights can be steep. The original Napster, a peer-to-peer file sharing network, allowed people to share MP3 files, meaning that one person would purchase a song file and then share it with a large number of people, who could download copyrighted music for free. Recording artists and their labels were upset at the free distribution of their work by Napster, who was earning money from the platform that allowed the free distribution. Napster was taken to court by several recording companies and found guilty of contributing to copyright infringement.[13] Napster "was forced to shut down the site after making a public apology and paying $26m in damages."[14] Music sampling, which makes great music, is subject to copyright, and musicians who sample without permission can be found guilty of copyright infringement.

Another intellectual property protection that you will likely encounter someday, if you have not already, is a **nondisclosure agreement.** A nondisclosure agreement is a legally binding contract that identifies confidential information that the parties involved in the agreement will share with each other, but which they may not share with others. Everyday examples of unilateral nondisclosure agreements for which you do not need to sign a contract but you still participate in are professor–college student, doctor–patient, priest–penitent, banker–client, and attorney–client. Unilateral means that while two parties are involved in the agreement, one person discloses the information (college student, patient, penitent, client) and the other party maintains confidentiality (professor, doctor, priest, banker, attorney). There are also mutual agreements in which all parties are expected to maintain confidentiality about the information identified in the agreement. If you intern or work for a company that wants to protect its trade secrets, you may very well be asked to sign a nondisclosure agreement. If you are asked to sign a nondisclosure agreement, make sure that you fully understand why you're being asked to sign one, the details of the agreement (what you are agreeing to), and the consequences of breaking the agreement. Have your questions or concerns addressed before you sign it.

The bottom line is that in technical writing, as in all intellectual pursuits, stealing ideas or the expression of ideas from others is not ethical and is absolutely unacceptable. It does not matter that "everyone" seems to be doing it, or that online resources make locating and using such material a trivial task. If you use the ideas or the expression of ideas (words, images, sounds, photographs, etc.) of others without acknowledging the source, you are stealing from the author, you are stealing from those students who are doing their own work, and you are stealing from yourself by turning what could be a valuable educational experience into an exercise in thievery. Also, if you commit plagiarism, especially in the professional world, you might find yourself on the wrong side of copyright, patent, trade secret, or trademark protections. That could subject you and your employer to severe criminal and civil sanctions, which, at the very least, would not be a career enhancement.

Learn to do it right the first time while in school. Document all sources with citations or notes at the point in the paper where the materials are used, and include a complete list of sources, usually at the end of the paper. See Chapter 16 for a more complete discussion of documentation and documentation requirements.

Image Alteration and Ethics

Without a doubt, digital image processing lends itself to abuse and misuse. The relative ease, availability, and effectiveness of hardware and software, combined with the inherent credibility of photographic images, provide a powerful tool for misrepresenting reality. As previously discussed, modern technology provides almost anyone with the power to steal or plagiarize intellectual property in digital form. Additionally, this same technology provides technical writers with the means to modify material with the intent of misleading the reader. Concerning the ethics of image alteration, the distinction between the intent to do good or to deceive applies.

This is especially true in the case of altered photographs. Technical writing always must be accurate and correct, and serious ethical concerns exist regarding accuracy and correctness when images are edited. Chapter 17 provides a discussion of image alteration and makes the point that modifying images is ethical or even desirable in technical writing when these modifications serve to clarify or complement—but not when they are used to misrepresent or deceive. If the image alteration has fundamentally changed the information it represents for the purpose at hand, that fact must be acknowledged. To avoid any ethical questions when you are using such altered images, always indicate to your reader that the images are altered. The easiest way is to acknowledge in the image's title and caption that it is, in fact, an edited image and then to briefly describe the nature and purpose of the alteration.

Social Media Presence

Using communication skills and resources with the intention of doing good extends to social media presence. Students and employees are de facto representatives of their schools and places of work, and their public communications reflect on their school or employer. This is why news of employees or students behaving rudely or offensively is of concern to organizations, institutions, and businesses.

At the writing of this book, employment in the United States (with the exception of the state of Montana) is "at will," meaning that employers can—except for firing someone because of their gender, race, religion, or other protected status—terminate employees without reason. There are some exceptions. For example, most states have a policy exception that prevents employers from retaliating against an employee who follows public policy or refuses to do something that would go against public policy. There are also implied contract exceptions, implied-in-law exceptions, and statuary exceptions.[15] Personal communication outside of the workplace can have an effect at work. For example,

- **Personal TikTok**. An oncology nurse bragged that, in times of COVID-19, they don't wear a mask unless they're at work.[16] They were put on administrative leave.
- **Personal Tweet**. A communication director lost their job after making a racist tweet before boarding a plane to join their family on vacation.[17]
- **Personal Facebook post**. A daycare worker lost her job after complaining about how much she hates her job and being around kids.[18]
- **Personal Tweet on corporate account**. A contractor for Chrysler lost their job after posting an explicit tweet from the official Chrysler account.[19] Their firm also lost their account with Chrysler.
- **Personal Instagram post**. A marketing executive made a celebratory post about a new client—against company policy and the executive's contract—that only a small number of followers would see (~200).[20] They lost their job.

Things that you say online can cost you at work, and companies are starting to have policies to clarify what is and is not acceptable on social media in order to protect their organizations' brands.

It isn't just after you have gotten a job that you should be careful about what and how you communicate on social media. Employers check applicants' social media accounts, and employers use online search engines to learn more about the people applying for jobs at their organizations.[21] Be proactive and build a professional and ethical online persona. If you aren't already, be cognizant of how you use and conduct yourself on social media. Some things to avoid posting on social media include:

- Complaints about your job, supervisor, employer, customers, clients, classes, instructors, school
- Offensive content (racist, sexist, ageist, etc.)
- Profanity
- Illegal or sexual content
- Partying
- Information that reveals or announces things prematurely or inappropriately
- Information that isn't yours to announce

In summary, ethical considerations are entwined in the work that you do as a representative of your discipline and organization. When you communicate, your moral duty and obligation is to apply the power of technical communication to the purpose of doing good and worthy things. Be good and do good.

Plagiarism Exercise

Situation 1

You are enrolled in a technical writing course and have been assigned the task of writing a process description on the operation of a four-stroke, internal combustion engine. A close friend wrote a similar description for another technical writing course at a different university last year and earned an A. They offer you their paper as a gift, along with the original word computer file. They tell you to change the name, print it again, and turn it in. Since the original author has given you the paper and you now own it, is it plagiarism for you to do as they ask? After all, they argue, you are not stealing anything—the paper is now yours!

Situation 2

What if you actually wrote the paper that your close friend turned in under their name a year ago and for which they received an A? Now you are in a technical writing class at another university and have been given the same assignment. You still have the computer file for their paper on your hard drive. Since you were the original author, is it plagiarism if you change their name to yours and turn in the paper for credit? If so, do you even need their permission?

Image Alteration Exercise

The author and family wanted all three sons to be pictured together in front of a campus landmark on graduation day so that they could complete the family's scrapbook of a crazy year of COVID-19. The family members were the only people in the area and had been quarantined together for two weeks.

For the following two hypothetical situations, is the use of the altered image ethical? Why or why not? Explain your reasoning.

Situation 1

The author thought the image might make a great example of how people should deport themselves on campus to comply with the university's guidelines for keeping students, faculty, and staff safe and healthy. The author shows Figure 2.2(a) to students in the department's orientation class as an example of how students used to stand before COVID-19 and how quarantined groups can stand outdoors, and Figure 2.2(b) as an example of how students should stand post-COVID-19 (or until the vaccines effectively prevent the spread of COVID-19).

(a) (b)

Figure 2.2(a–b)
2.2(a) is the original photo taken by the author. 2.2(b) is an altered version of the original photograph.
Leslie Potter

Situation 2

The author is meeting with potential students and parents about their department and the university. As part of the discussion, the parents express concern about their student's safety on campus and the university's COVID-19 guidelines, so the author shows them Figure 2.2(b) as evidence that all students (and their families) are following COVID-19 guidelines.

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CHAPTER THREE

03

00011

Note-taking

Imagine that your job is to understand and manipulate hundreds of information sources efficiently and effectively. The information includes subcontractors, their locations, products, logistics of getting products to the main production facility, schedules, contract limitations, impending weather constraints (blizzard or hurricane, anyone?), and more. Could that ever happen? Absolutely! Is it intimidating? It does not have to be, and it will not be with a good understanding of how to herd the information.

What Is Note-taking?

Note-taking is the act of recording information—whether writing, typing, voice-recording, or something else—from a variety of sources into a single location, usually a text or document. In general, everyone understands the idea of what it means to take notes because everyone has done it . . . maybe in a class, at the doctor's office, or when your parents are reminding you of all the things you need to do before you move to college. Sometimes we rely on mental note-taking. In short-term or short-list situations, this can work well; we are likely to remember and act as needed. Oftentimes in our technical jobs, though, we need to document our notes. There are several reasons for this.

As individuals, we

1. must be able to remember a lot of information efficiently and
2. might need to access the information over long periods of time.

As part of a team, we

1. want to contribute to the institutional knowledge base and
2. must be sure that our actions are ethical and legal, including both the notes we take and do not take.

To be effective and efficient note-takers, we must understand why and have a strategic plan for taking them. Let us return to the opening scenario of this chapter, the one where we are trying to keep track of and herd hundreds of sources of information. We take our cue from Border Collies—naturals at quickly processing and sorting information. Perhaps you have seen them amaze in agility like *Pink* at the Westminster Dog Show.[1] Or maybe you have seen Border Collies in herding competitions like *Skid* in the Scottish National Herding trials.[2]

Let us check in with Skid, whose job is to herd sheep . . . lots of them! On any given day, his job is to literally move hundreds of sheep from Point A to Points B and C efficiently and safely. This move requires absorbing copious amounts of information, including understanding what Skid's farmer wants, what each sheep is doing and will be doing, the terrain, ambient conditions, the physical constraints of fences and gates, and a whole lot of other sensory data. For most, this results in sensory overload . . . game over! But for Border Collies, this is all in a day's work. They are successful at it because they can take in all this information, pare it down quickly to key points, and make good decisions based on their mental notes. Their farmer aids this process by reducing very complex and intricate commands into notes as well. When Skid's farmer is thinking, "Skid, I want you to round up half of those sheep and move them from the pasture through the north gate and into the corral, and put the other half in the south paddock" but says, "Skid, come-bye, walk, in here" and gestures or whistles, Skid understands instantly and processes the information with a quick assessment . . . in other words, Skid takes good notes (see Figure 3.1).

Border Collies are born with a herding instinct. They want to do their jobs. They know WHY—because when they do their job well, it comes with treats and praise. But they still must be taught what to do and how to do it. People are much the same. While we all know how to write words on paper or type on a computer, we benefit from training in what to do and how to do it. People also benefit when they understand WHY as well.

A quick search of the internet for "good note-taking" returns approximately three billion hits in less than one second. There are several different standard processes for good note-taking, including but not limited to

1. Outline
2. Mapping
3. Sketch-noting
4. Cornell

You are probably familiar with outlining—this works well if you know ahead of time what the agenda will be. It is harder to use this system in real time if you are trying to process new information without any prior knowledge of what will be covered. It is also a little overwhelming to go back and review these notes later as there is not any intuitive summarization beyond the notes themselves.

Mapping involves connecting ideas together. You can draw a main idea "cloud" and connect nodes of ideas to it. You can also connect cloud to cloud. If you are a visual person, this might be a good system. Another good system for visual notes is sketch-noting (see Figure 3.2). One drawback to sketch-noting is that it is difficult to be thorough in real time.

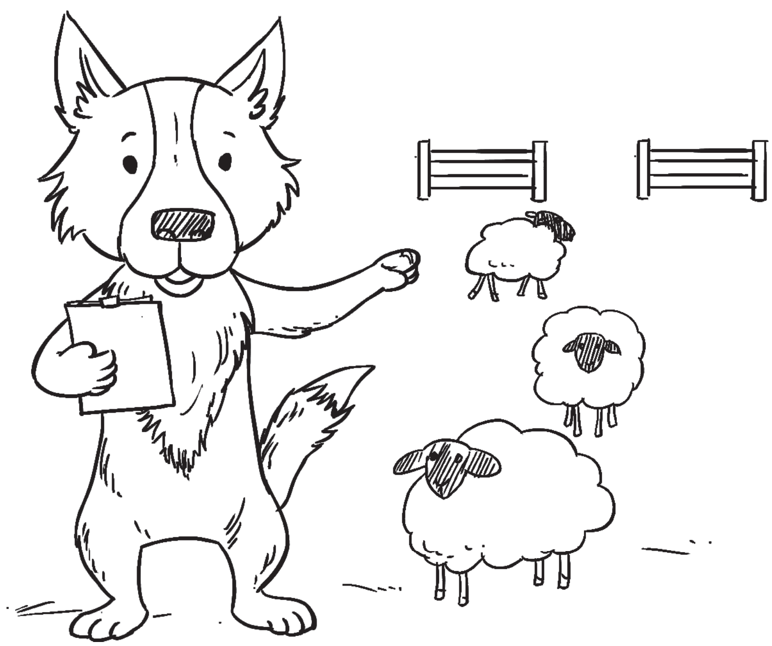

Figure 3.1
Usually people take notes for themselves, with the purposes of learning, documentation, and future reference. In these cases, you are your own audience, and you will know your purpose and the context of the situation. Sometimes, however, we take notes for others or potentially for others to reference. In these cases, consider what your audience needs and why.

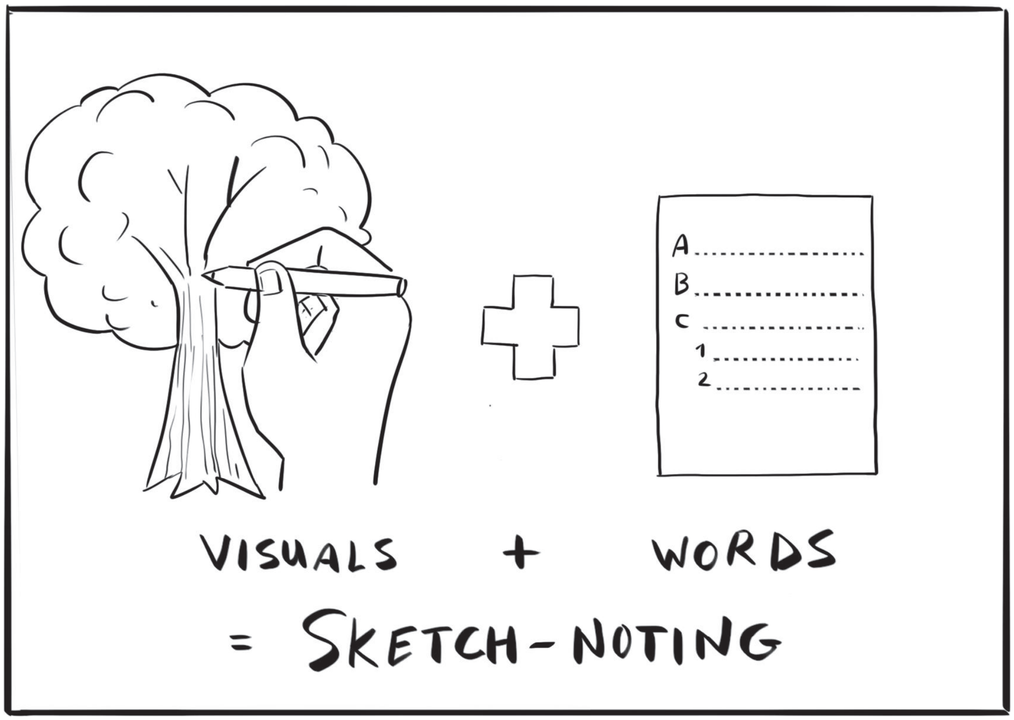

Figure 3.2
Sketch-noting is the process of using both visuals and text to conceptualize ideas, connect those ideas, and add details.

The Cornell method of note-taking is a great system for taking notes in real time, to which you can add visual cues later. The essence of the Cornell system is to divide your document into four sections as seen in Figure 3.3. A title at the top provides the summary topic for the entire set of notes, while the summary

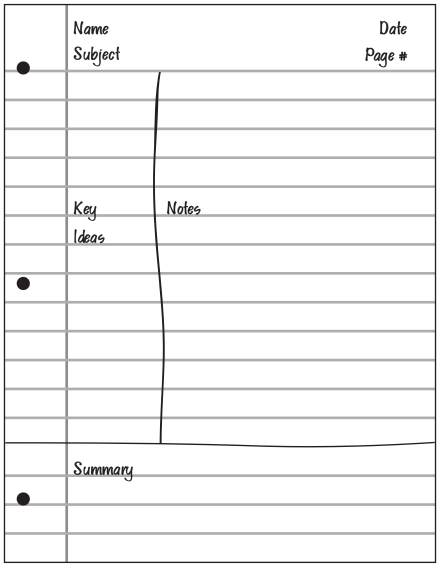

Figure 3.3
Cornell note-taking is a process that facilitates organized and easily referenced notes.

section at the bottom provides a summary for the page. The right side of the document is for notes taken in real time, and the left side of the page is for key ideas. Sketches of information or key ideas can be included if the visuals help you quickly reference information.[3]

With any note-taking system, it is important to capture citation information as you go. It is painfully inefficient and often ineffective to try to remember where you found information days or even hours later. You should document source information including websites, access dates, authors, titles, page numbers, and anything else that could be required by the style guide you are using.

Our focus in this chapter is to remind you of some of the "how" options, but also to have you understand the WHY you want to be good at taking notes. There are three very critical reasons, including the utility of the exercise, the ethical implications, and the potential legal ramifications of note-taking.

Utility

The science is clear; taking notes helps you retain information more effectively and efficiently.[4] Note-taking forces your brain to think about things, and the result is that you are more likely to store the information in long-term memory. More complete notes are also better for learning than brief notes, so there is a balancing act between too much and not enough.

That said, one thing you will never see a Border Collie do is take notes about sheep on a computer, and it turns out that if your purpose for taking notes is to learn, you should follow their lead. There are many sources and studies which concur: using a computer to take notes is not as effective for learning as writing on paper.[5,6]

Ethical Implications

Recognition of the importance of ethics is pervasive in science and engineering fields, and ethics includes taking notes. The National Society of Professional Engineers (NSPE), for example, has numerous case studies involving the ethics of note-taking in specific situations. If an engineer has information in their notes, is it ethical to report them or unethical to report them? You might not be surprised to read that "it depends," and that it is your responsibility to know what "it depends" means for your discipline, field, position, and the situation. The important takeaway is that note-taking matters, and as a professional, you should investigate the ethics of your professional note-taking.

One example of a real NSPE case involving the ethics of note-taking is titled "Duty to Report Unrelated Information Observed During Rendering of Services." It involved an engineer who discovered information pertinent to a preexisting bridge defect that might have contributed to a death. The engineer documented what they saw in their notes, but was then asked by a public agency to not include that finding in a final report. While the case requires extensive background context and has multiple findings, among the findings, the NSPE Board determined that the engineer did the appropriate thing by documenting the information for possible future reference as appropriate.[7]

Another case from NSPE, titled "Employment–Employee/Employer Files," states that

> NSPE Code Section III.9.d. makes it clear that an engineer's designs, data, records, and notes referring exclusively to an employer's work are the employer's property and not the property of the engineer.[8]

Notes are an important part of learning, recalling, documenting, and justifying information in technical jobs. So maybe you are now thinking, "Okay, my job is my job . . . but that doesn't affect me personally, right?" Wrong. Potentially affecting us on a very personal level, studies show that jurors who take notes during trials perform more appropriately for the cases for which they serve. In one study called "Effects of

Notetaking on Verdicts and Evidence Processing in a Civil Trial," some interesting observations include that the more competent jurors took notes during the trial, and that jurors who took notes had better recall of information than those who did not take notes.[9] If and when you serve the public as a juror, do you want to do the best job possible?

Legal Implications

If the ethics of note-taking are not enough incentive, consider the potential legal ramifications. You or your company can be held legally responsible if you know something that is detrimental to the safety or welfare of others and do not share it with the appropriate parties, or if you knew something and didn't record it in an attempt to not have a written record. According to Christopher Proskey, patent attorney and industrial engineer,[10] "notes are often important documents in litigation, as they represent the closest thing you can get to what a witness was thinking at the time."

> Every litigation begins with what is called the "discovery process," when each party tries to determine what the other party knows about the subject of the litigation. One step in the discovery process is sending the other side "Requests for Production of Documents" (RPDs). RPDs demand the other side to produce all documents that are relevant to whatever topic is specified in the RPD. Each party that receives an RPD has a legal obligation to provide the requested documents. Often notes are relevant to an RPD and therefore the notes must be produced to the other side—irrespective of whether the notes in question are good or bad for the witness or the side they are on. If the requested party fails to provide relevant documents in their possession, they are breaking the law (which means they can go to jail).
>
> Notes produced in response to an RPD often provide the other side with valuable insight and information. These notes can then be used to test witnesses' opinions or recollection during depositions or at trial, and they can be used as evidence in motions or at trial. Many witnesses have been placed in uncomfortable positions on cross-examination when they say something very different from their notes taken previously. For these reasons, notes have a substantial impact on the outcome of many cases.

We see similar concerns again with the NSPE regarding the failure to share information. In Case 95-5, called "Failure to Include Information in Engineering Report," an engineer chose not to report all of the technical information available and relevant to the situation. The findings reiterate that the NSPE Code of Ethics requires that engineers "shall include all relevant and pertinent information in such reports, statements or testimony."[11] Mr. Proskey continues:

> Note-taking also has particular importance in patent law. For many years, the U.S. was under the "First to Invent" rule, which meant that the first person to "invent" something was awarded the patent for it. This resulted in many litigations where Party A filed a patent first, but Party B challenged them saying they invented it first, and therefore Party B should own the patent instead of Party A. This fight often came down to a battle of the notes. Essentially, whoever could prove, using their notes, that they invented it first, was awarded the patent.
>
> The law changed with the Leahy-Smith America Invents Act of 2012, which changed the law to the "First to File" rule. This new law means that the first party to file a patent is awarded the patent. However, this change did not make notes any less important in patent law. Following the Leahy-Smith America Invents Act, inventor notes can be very useful in litigation as a defense against a claim for patent infringement.
>
> For example, if Party A files a patent and accuses Party B of infringing their patent, Party B can try to invalidate Party A's patent by proving Party B was using it first. If Party B can prove that Party B was using the subject of Party A's patent before Party A filed the patent, Party B can claim Party A's patent is invalid for being in public use or available to the public before Party A applied for the patent. Inventor notes are often critical in this proof.

Digital Footprints

While most of this chapter is about the notes that **you** will take, we would be remiss if we did not address digital footprints, which are the "notes" that others take **about** you.[12] Whether you know it or not, every click of a mouse and every keystroke are being saved for posterity when you are working on the internet. If you save to a cloud-based server, your files are out there forever, regardless of "deleting" them. If you (responsibly!) back up your work to a remote back-up service, it is there forever. If you post to social media sites (e.g., LinkedIn, Snapchat, Instagram, YouTube, Facebook, Twitter, TikTok, etc.), your digital presence continues to expand and, theoretically, can be found and used to define you. "But I deleted the file! But I clicked on the wrong link! But I didn't mean to post it publicly!" None of those matter. You must remember that your digital footprint is forever and could potentially be accessed by anyone. Consider that carefully before you start typing.

That said, there might be situations where you will appropriately comment within a digital platform or file, but then inappropriately share that information. What happens then? The NSPE addresses an example of that as well. In Case 09-11, titled "Discovering Embedded Comments in Electronic Documents Damaging to Adversary," they note that the ethics and legalities of some situations overlap, and that in such cases, the legal ethics must be addressed first.[13]

In summary, note-taking is a critically useful tool for how we do our jobs and live our lives, but we want to make good, informed decisions. Knowing what system works best for a given situation, understanding what to document and when, and being cognizant of the permanence of our notes are all critical to being successful.

Note-taking Checklist ☑

- ☐ Should we take notes?
- ☐ What method of notes works best for us and the situation at hand?
- ☐ Will these notes be the property of my company or are they personal?
- ☐ If using the Cornell system, do I have templates pre-drawn?
- ☐ If using the outline system, do we know any guiding information ahead of time?
- ☐ How will we store/maintain/disseminate our notes?

Exercise

1. Create a template for Cornell note-taking, go back to the start of this chapter, and take notes. Can you summarize?
2. Watch the video on Sketch-noting and create notes for it: https://www.youtube.com/watch?v=pZgMpjjgCRA&feature=emb_logo.
3. When you take notes in a class, who is the audience (for the notes)? What is the purpose?
4. When you take notes while meeting with a client, who is the audience (for the notes)? What is the purpose?

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CHAPTER FOUR

04

00100

Technical Definition

There are many well-understood powerhouses in this world: the hard-working, no-questions-asked, get-the-job-done kinds of creatures and things. They are not fancy, and they do not subscribe to needing recognition, but they are dependable and solid. Think of your toolbox—perhaps your trusty hammer comes to mind. After a snowfall, you grab that same old shovel you have had for ten years. In your kitchen, your can opener never fails. Likewise, the working dogs include many breeds that are just like this: solid, simple, and straightforward. The Doberman Pinscher's job is to announce change. None do it better, quickly and easily convincing all that they mean business (see Figure 4.1). The Samoyed is built to pull sleds in −60° F weather. They manage this easily with their beautiful fur coats. The Newfoundland is literally born to swim, with webbed feet that allow them to haul in fishing nets and specialize in water rescues. Working dogs get the job done and do not ask for appreciation. In technical writing, there are some working dogs. The first of these, called the technical definition, helps people understand the meaning of words: not a fancy job, but a very important one.

Virtually any kind of technical writing includes one or more technical definitions, whether it is a formal technical report or an informal presentation. Technical writers must be able to effectively define terms for their audience, no matter their audience's level of knowledge. In fact, much of a technical writer's job will be to write for non-expert audiences with little to no knowledge or experience with the subject.

Figure 4.1
The working dogs of technical writing, represented here by the Doberman Pinscher, include technical definitions, descriptions of mechanisms, and descriptions of processes.

What Is a Technical Definition?

In technical writing, **definition** is the process by which one assigns a precise meaning to a term. If the goal of technical writing is to reduce abstraction and avoid (mis)interpretation, the goal of writing a technical description should be to define a term so well that there is only one way to understand what the term means. To define a term, it must be placed into a classification and then differentiated from other terms in that same classification. Technical definitions are relatively easy to write, except for some pitfalls that will be addressed later. The format for a technical definition is straightforward and works like this:

Term = *Classification* + Differentiation

For example, if a writer were to define the stall condition that an airplane experiences when it loses lift, they could start with the term **stall,** add a classification—*flight condition*—and then differentiate it from all other flight conditions, in this case, by a stall's unique characteristics. The definition might read something like this:

A **stall** is a *flight condition* in which the lift produced becomes less than the weight of the airplane, and the airplane stops flying.

That seems simple enough; but what happens when a term, such as **stall,** has multiple definitions in many contexts? For example, in a sports context, **stall** can mean a tactic to drain time on the clock and in an agricultural context, a **stall** is a compartment in a stable or barn for a domesticated animal (think horse or goat). In such cases, it might be necessary to add a qualifier in front of the definition statement to supply the necessary context. The qualifier is important when the general context for a definition needs to be established up front. If the context is known or is obvious, a qualifier is unnecessary. For example, in an aeronautics study on aircraft wing design, the context of **stall** is obvious. It is clear that **stall** in this case has to do more with the loss of lift than with, say, a defensive maneuver employed by a basketball team to slow down the pace of the game or where a horse stays overnight.

When context is needed, the format for the definition is

(Qualifier +) **Term** = *Classification* + Differentiation

Note: The parentheses and boldface have been added here and earlier for clarity. Visually emphasizing the term using boldface or italics, however, is a good practice when writing any definition.

Look at Figure 4.2, which provides three definitions of the same term in different contexts. Notice how the term **stall** has three totally different meanings depending on the context, and how each definition begins with a qualifier that makes the context clear from the start.

In the first example, **stall** refers to a car that has suddenly stopped running. This condition typically occurs in the middle of heavy traffic, in bad weather, or when you are desperately trying to make a flight for which you have nonrefundable tickets.

In the second example, **stall** refers to what happens when an airplane does not go fast enough to stay in the air. Pilots routinely stall their airplanes right above the runway when landing them, in which case stalling is good. Sometimes, however, they stall them inadvertently in situations when they really want their planes to keep flying, in which case stalling is bad.

The third example of **stall** relates to farms, barns, stables, and livestock. The authors have had their fair share of experiences with stables and barns—though, hopefully, they have grown past some early childhood observations that they "were raised in a barn."

(qualifier) **term** *classification*

(In driving), a **stall** is an *operating condition* in which a sudden and unexpected loss of power occurs.

(In flying), a **stall** is a *flight condition* in which the lift produced is less than the weight of the airplane.

(In farming), a **stall** is a *single, three-sided compartment* that is designed to shelter an animal in a barn or stable.

Figure 4.2
The **term** stall has multiple meanings, three of which are defined here using the format for technical definitions: (Qualifier +) Term = *Classification* + Differentiation.

Classifications and Classes

Often, the most difficult part of writing a technical definition lies in determining the proper classification for the term. The class should be a general category in which the term fits, but it cannot be too general. For example, consider that 33-kilohm, 1-watt carbon resistor from the abstraction funnel in Chapter 1. In the following sentence, the term is defined by using the very generic classification of *device.*

The **33-kilohm, 1-watt carbon resistor** is a *device* that impedes the flow of electric current.

The problem with this classification is that *device* could mean all kinds of different things, most of which have nothing to do with circuit components; consequently, its inclusion does not really help specify the meaning of the term. By changing *device* to *circuit component,* however, the meaning can be narrowed considerably for the reader, even before the classification is differentiated.

One trick for classifying a term is to build an abstraction funnel and use the filter one abstraction level above the term. In the abstraction funnel shown in Figure 4.3, moving up a filter level from the term *33-kilohm, 1-watt carbon resistor* to the term *resistor* is not an option because the classification would be derived from the original term. That would yield the following circular definition:

The **33-kilohm, 1-watt carbon resistor** is a *resistor* that impedes the flow of electric current.

Because the 33-kilohm, 1-watt resistor is obviously a resistor, this classification does not help define the term. In fact, it contributes nothing other than useless circularity. If possible, try to find a classification that is not derived from the term. In this case, the easiest solution is to just move up the abstraction funnel one more filter. *Circuit component* helps to define the term and has no circularity with the term.

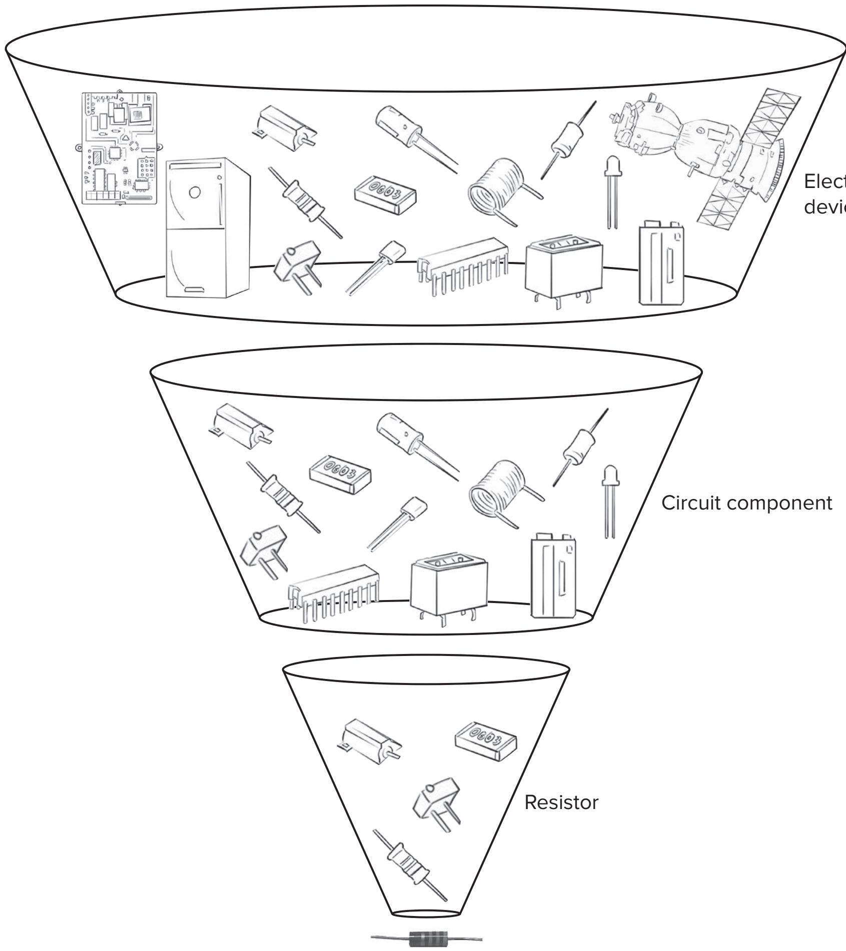

Figure 4.3
A funnel of abstraction. The high-level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.

Differentiation

The next step in defining the term is to differentiate it from all the other members of its class. Differentiation involves narrowing the meaning of the term to just one possibility within the class. Clearly, it would be easier to narrow the class of "circuit components" to a particular resistor than it would be to narrow the class of "device" to a particular resistor. There are all kinds of devices in the world and relatively few circuit components.

The class "circuit components," however, still contains many possibilities: capacitors, diodes, switches, potentiometers, inductors, transistors, and IC chips, to name a few. In this case, a good approach is to focus on the function of a resistor, which is to impede the flow of electric current, and to use that function to differentiate the class. Doing so yields the following definition:

> The **33-kilohm, 1-watt resistor** is a *circuit component* that impedes the flow of electric current.

Avoiding Imprecision

In writing technical definitions, it is easy to do something that is "not good." For example, when we are defining the common computer term **hard drive,** it is probably best to add a qualifier unless the context of computing is obvious. (There are other kinds of hard drives, such as one through the Mojave Desert in July without air conditioning.)

So how about this definition?

> In computing, a **hard drive** is an *input/output device* for the nonvolatile storage and retrieval of data.

Think about this one for a minute. If the reader does not know what a **hard drive** is, what is the likelihood that the reader will know what an *input/output device* is, much less the meaning of nonvolatile storage and retrieval?

How about this definition of a water softener's resin?

> In water softening, **zeolite** is an *exchange resin* that gains sodium ions while releasing calcium and magnesium ions.

This definition is fine as long as you are sure your audience will know what *ions* are, not to mention *exchange resin, calcium,* and *sodium.* An audience familiar with water softeners and basic chemistry certainly would, but what about a less technical audience—perhaps a nontechnical manager who controls your funding? For a less technical audience, we might rewrite this definition as follows:

> In water softening, **zeolite resin** is a *special material* that releases sodium into the water while absorbing calcium and magnesium from the water.

Notice that there is no mention of ions or ionic exchange. A nontechnical audience would not know (or probably even care) about such things as ionic charge and exchange.

Extensions

You must always consider the reader's knowledge and skill level when defining terms and concepts. However, at times you may have no choice but to violate the principle to achieve your goal. Sometimes a simple term or concept that the audience will understand is just not available. In fact, sometimes the only thing to do is to define the term as best as you can, then add extensions to your definition to clarify your meaning.

Extensions can take many forms. In the following examples of the most common types of extensions, the original definition is included in brackets followed by the extension.

Further Definition

Use further definition when you need to define terms you used in your original definition.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite is a collection of small polystyrene beads that forms a resin carrying a negative charge.

Comparison and Contrast

Use comparison and contrast when you need to show differences or similarities.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite collects calcium and magnesium ions in the same way a feather broom collects dust.

The comparison or contrast can be with a concept your audience is familiar with, which can make understanding a new concept easier.

Classification

Use classification when you need to organize information into categories.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite is used for ion-exchange "home" water softening, while lime and soda ash are used in precipitating municipal water softeners.

Much like comparison and contrast, classification can help your audience by connecting to concepts with which they are familiar.

Cause and Effect

Use cause and effect when you need to demonstrate why something happens or when you need to trace results.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite carries a negative charge and attracts the positively charged calcium and magnesium ions in water.

Process

Use process when you need to list the steps of a procedure.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] First, the zeolite is charged with sodium ions from a salt brine solution in the regeneration step. Once charged, the zeolite absorbs calcium and magnesium ions from passing hard water while releasing sodium ions into the resulting soft water.

Exemplification

Use exemplification when you need to give real or analogous examples.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Examples of the zeolite group include analcimes, chabazites, and gismondines.

Etymology

Use etymology to show the linguistic genesis of the term.

> [In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] The term **zeolite** comes from the Greek words *zeon,* meaning "to boil," and *lithos,* meaning "stone." It was first coined by the Swedish mineralogist Alex Fredrick Cronstedt in the 18th century.

Required Imprecision

In some situations, you might need to sacrifice desired precision in your definition to achieve the required level of communication. At times, for example, it is foolish to attempt to achieve expert-level precision with a non-expert audience.

Consider the following two definitions of a **black hole**—the astrophysical phenomenon that exists in space.

Definition 1

In astrophysics, a **black hole** is a set of events from which it is not possible to escape to a large distance. A black hole gets its name from its boundary, called an **event horizon,** which is formed by the paths in space-time of rays of light that just fail to get away, hovering instead forever on the edge and, consequently, moving on paths parallel to or away from one another.

Definition 2

In astrophysics, a **black hole** is a collapsed neutron star whose gravity is so great that even light cannot escape. Although fusion reactions within this collapsed star still may emit brilliant rays of light, when it is viewed from the outside, the black hole appears to be a totally dark void in space.

Definition 1 functions at the expert level. For theoretical physicists, it provides a precise and accurate description of a black hole and thus is appropriate for their needs. But for the less knowledgeable, reading it represents a mind-twisting experience that, in many cases, can leave readers more confused about the term than they were before they read it.

Definition 2 functions at the level of the average reader. It is not nearly as precise or accurate as the first definition, but it communicates the basic gist of what constitutes a black hole in space.

The problem is that to be precisely correct and absolutely accurate, you would have to function at a level where you cannot effectively communicate with the average reader. If you were writing for the average reader, you would have to make a decision here: either be absolutely correct and communicate less effectively, or be less than absolutely correct and communicate more effectively. Since your goal is effective communication, your decision is obvious.

A Word about Defining Specifications and Standards

Defining specifications and standards is a specialized activity. Such documents can take many different forms, depending on what areas of engineering and science are involved and whether the documents are designed to meet commercial, industrial, or government standards. Writing these kinds of definitions is far more complex than simply defining terms.

Specification documents precisely state particulars, including requirements, designs, implementations, and testing. In engineering and science, specifications normally involve goods and services being developed under some type of contractual obligation. The specification precisely defines the quality of work and performance standards required by the contract. In addition, as will be shown in Chapter 5, specifications can describe mechanisms in terms of their physical and functional attributes.

Specifications also form the basis of technical standards. Standards are accepted or established methods, measures, or designs for accomplishing specific tasks. Standards exist for everything from data transfer protocols and cable connectors to air conditioning coolants and drinking water.

Definition Checklist ☑

- ☐ Have I fully analyzed the purpose of my project, and do I understand the skill and knowledge level of the audience?
- ☐ Have I defined the term by first classifying it in a way that adds precision and understanding for my audience and that serves my purpose?

- ☐ Have I differentiated this classification to distinguish this term from other members of its class?
- ☐ Have I determined whether the context is clear and, if not, whether it is critical to the definition? If the context is unclear and critical, have I used a qualifier before the term?
- ☐ Have I avoided defining a term using the same term?
- ☐ Have I avoided using terms that themselves need to be defined? If not, have I explained these terms with extensions?
- ☐ Have I chosen extensions to my definitions that are appropriate for my audience and purpose?
- ☐ Have I compromised my fundamental purpose (communicating with the reader) by including inappropriate or irrelevant information or precision?

Exercise

Read each of the following definitions and try to determine the context, audience level, accuracy, and purpose for which they were written.

- A **resistor** is a small electronic part that reduces the amount of electricity flowing through a circuit.
- A **resistor** is a circuit component that converts electrical energy to thermal energy and, in the process, determines the current produced by a given difference of potential.
- **Resonance** is a systemic condition in which small amplitudes of a periodic agent produce large amplitudes of oscillation or vibration.
- **Resonance** is a natural means of amplification that makes a musician's horn sound louder.
- **Ionization** is the electrostatic process by which a neutral atom or molecule loses or gains electrons, thereby acquiring a net charge.
- **Ionization** is the phenomenon that creates lightning in thunderstorms.
- **Ergonomics** is the field of study by which we make machines easier to use.
- **Ergonomics** is the systematic consideration of physical, psychological, and social characteristics of human beings in the design of tools and equipment, the workplace, and the job itself.

CHAPTER FIVE

05

00101

Description of a Mechanism

Much like technical writing typically includes one or more technical definitions, almost every kind of technical writing includes a technical description of either a mechanism (Chapter 5) or a process (Chapter 6). Whether it is a formal technical report, a proposal, a refereed article, or an informal presentation, you may need to include a technical description. There is no standard length for technical descriptions because they are what the audience needs them to be. Technical descriptions can be a paragraph within a proposal, a couple of paragraphs in a technical report, or even a stand-alone document of several pages.

Technology involves physical devices called **mechanisms**. Being able to describe mechanisms precisely and accurately, and at a level and in a way that the reader needs and can understand, is perhaps the most essential component skill of technical writing. This skill is particularly important for those who produce documents involving specifications and instructions.

Just like technical definitions, when it comes to descriptions of mechanisms, we have another working dog. While at first glance you might not fully appreciate them, they are no-nonsense, elegant, strong, and powerful (see Figure 5.1). Their grace and beauty show through when your audience fully understands the technical mechanism you are describing with minimal to no effort.

Figure 5.1
The Samoyed is another working dog whose ability to get the job done is sometimes underappreciated. Note that the dogs are pulling a sled and wearing harnesses, which are mechanisms that could be described.

What Is a Description of a Mechanism?

Descriptions of mechanisms are precise portrayals of material devices with two or more parts that function together to do something. Mechanisms can range in complexity from circuit components and mechanical fasteners, to supercomputers and space shuttles, to a bike that parents must assemble for a birthday.

In Figure 5.1, the Samoyed dogs are pulling a sled, which is how the breed has earned their keep in Siberia for centuries. The sled is a mechanism. So are the harnesses that the dogs wear. How would you explain the technical requirements for either of these to an audience that is considering making or using them? As you might notice in Outline 5.1, the primary focus in writing a mechanism description is on the physical characteristics or attributes of a device and its parts. Mechanism description documents are built around precise descriptions of size, shape, color, finish, texture, and material, and they can be written for a wide range of audience skills. Such documents also normally include figures, diagrams, or photographs that help to clarify the worded description. Some descriptions (e.g., components for space systems) might require additional attributes such as weight or mass.

Know Your Audience

If you are describing a mechanism (like a bicycle kickstand) for a patent application, this has a very different audience from the bike distributor reading a sales brochure or the frazzled parent after an 8-year-old's birthday party reading a set of assembly instructions. See Chapter 3 for a teaser on legal implications associated with patent law, but know that you will want to consult a patent attorney if your audience is at that level of need.

Outline 5.1 Description of a Mechanism

Introduction

- Define the mechanism with a technical definition (see Chapter 4), and add extensions to discuss any theory or principles necessary for the reader to understand what you are saying. Always make sure you add only what the reader needs for the purpose at hand. If the reader does not need any theory or operating principles, do not provide any. Consider any relevant context for the situation as well.
- Describe the mechanism's overall function or purpose.
- Describe the mechanism's overall appearance in terms of its relevant attributes—e.g., shape, color, material, finish, texture, mass/weight, and/or size.
- List the mechanism's parts in the order in which you plan to describe them.

Discussion

- ***Part 1***
  - Define the first part with a technical definition, adding extensions as needed to deal with theory or operating principles.
  - Describe the part's overall function or purpose.
  - Describe the part's shape, color, material, finish, texture, mass/weight, and/or size (as well as any other physical attributes appropriate for the mechanism and its function), using precise measures and descriptors. Also, be sure to use figures, diagrams, and photographs as necessary.
  - Transition from this part to the next part.
- ***Parts 2–n***
  - For each remaining part, repeat the pattern of defining, describing, and transitioning established for Part 1.

Conclusion

- Briefly summarize the mechanism's function and relist the parts described.
- Give a sense of finality to the paper.

A variation of the traditional mechanism description extends the treatment from just physical attributes to include a more expansive discussion of function and theory of operation. The organization of this type of mechanism description is still based on the parts, but the discussion of function and theory may be extended.

A more specialized mechanism description, the functional mechanism description, describes a mechanism with key physical attributes related to operating parameters. This type of description, which often includes specific operational standards, can be used to provide specifications for the mechanism. Technical specifications are normally written for an expert audience that actually needs to design, maintain, use, or evaluate the mechanism. For example, if you were the design engineer for the skis on a dog sled, you would need to include the material specifications (what kind of steel, how hard must it be, what must the surface finish be, how should it be formed, etc.). Your description should be fully understandable by the engineers who must make the product.

Moving from an Outline to a Complete Description

Outline 5.1 provides a model for a general mechanism description. To show how this model works, we will use it to describe a relatively simple mechanism: the 33-kilohm, 1-watt carbon resistor discussed in Chapters 1 and 4. The resistor is a relatively simple mechanism; most mechanisms are much more complex. Assume, for the purpose of illustration, that this description is for the average technical reader who requires only a general description of the resistor.

Example 5.1 General Mechanism Description

Description of a 33-Kilohm, 1-Watt Carbon Resistor

Introduction

The **33-kilohm, 1-watt carbon resistor** is a circuit component that impedes the flow of electric current. The resistor impedes the flow of current by converting a portion of the electrical energy flowing through it into thermal energy, or heat. This particular resistor can safely convert 1 watt of electrical energy into heat.

The 33-kilohm, 1-watt carbon resistor looks like a small cylinder with wire leads extending from each end. The casing's surface is composed of smooth, brown plastic with a shiny finish. Four equally spaced color bands (three orange, one gold) circumscribe the cylinder, starting at one end.

The resistor consists of the following parts: the carbon element, the wire leads, the casing, and the color bands (see Figure 5.2).

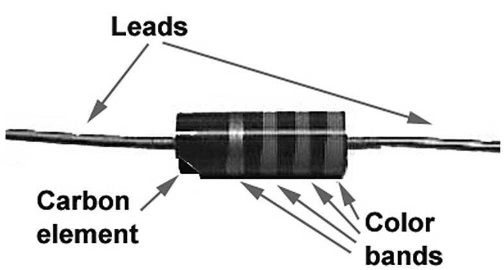

Figure 5.2
Parts of the resistor.